

5.1. CONVERGENCE AND RADIUS OF CONVERGENCE OF INFINITE SERIES

In this chapter, we shall develop methods of expanding a given function about a point a in powers of $z - a$. As we proceed, we recognize that this theory enables us in evaluating certain real and complex integrals easily.

We start with some definitions and results before we take up Taylor's and Laurent's series expansions. The following material is useful for the students to learn the subject and for more details the students are advised to study an exclusive Textbook on Complex Variables (for example : the theory of functions of Complex Variables—Shanti Narayan S. Chand & Co.).

Let $\sum z_n = z_1 + z_2 + z_3 + \dots + z_n + \dots$ be an infinite series of complex numbers. Let $z_n = a_n + ib_n$ where a_n, b_n are real. Then we can write

$$\sum z_n = (a_1 + ib_1) + (a_2 + ib_2) + (a_3 + ib_3) + \dots \quad \dots(1)$$

If the series of real numbers

$$\sum a_n = a_1 + a_2 + a_3 + \dots \text{ and } \sum b_n = b_1 + b_2 + b_3 + \dots$$

are both convergent (say) to A and B respectively we say that $\sum z_n$ converges to $A + iB$.

If $\sum z_n$ converges, as in the case of series of real numbers, we can prove that $\lim_{n \rightarrow \infty} z_n = 0$. The converse of this result is not true.

The series $\sum z_n$ is said to be absolutely convergent if $\sum |z_n|$ is convergent.

Let $\sum u_n(z)$ be an infinite series of functions $u_n(z)$ defined over a domain D . Define

$$S_n(z) = u_1(z) + u_2(z) + u_3(z) + \dots + u_n(z).$$

We say that the series $\sum u_n(z)$ converges to sum $S(z)$ pointwise if to $\epsilon > 0$ there corresponds a positive integer N depending on ϵ and z such that

$$|S_n(z) - S(z)| < \epsilon \text{ for all } n \geq N.$$

We say that the series $\sum u_n(z)$ converges to the sum $S(z)$ uniformly in the domain D if to each $\epsilon > 0$, there corresponds a positive integer N independent of z such that

$$|S_n(z) - S(z)| < \epsilon \text{ for all } n \geq N \text{ and for all } z \in D.$$

As in the case of series of real functions, here again, we have :

if there exists a convergent series of positive constants $\sum M_n$ such that

$$|u_n(z)| \leq M_n \text{ for all } z \in D \text{ and for all } n$$

then, $\sum u_n(z)$ uniformly converges in D .

(Weierstrass M test)

The following results are worth noting :

1. If $\sum u_n(z)$ uniformly converges in a domain D to the sum function $S(z)$ then $S(z)$ is continuous in D .
2. If $\sum u_n(z)$ uniformly converges to $S(z)$ in a domain D , then $\sum \int_c u_n(z) dz = \int_c S(z) dz$ assuming the existence of the integrals on the left side.

Definition. A series of the form $\sum a_n z^n$ is called a power series.

If $\sum a_n z^n$ converges at $z = z_1$ then it converges absolutely for all z such that $|z| < |z_1|$.

Definition. If $\sum a_n z^n$ converges for $|z| < R$ and diverges for $|z| > R$, then R is called the **radius of convergence** of the power series and $|z| = R$ is called the **circle of convergence** of the power series.

If R is the radius of convergence of the power series $\sum a_n z^n$, then the power series converges uniformly for $|z| \leq R_1 < R$.

The following is one of the most important theorems in complex variables theory.

5.2 Taylor's Theorem

[JNTU 2000, 2005S, April, Nov. 2006, Aug. 2007S, Nov. 2008S (Set No. 2)]

Let $f(z)$ be analytic at all points within a circle c_0 with centre at a and radius r . Then at each point z within c_0

$$f(z) = f(a) + f'(a)(z-a) + \frac{f''(a)}{2!}(z-a)^2 + \dots + \frac{f^n(a)}{n!}(z-a)^n + \dots \quad \dots(1)$$

i.e., the series on the right hand side in (1) converges to $f(z)$ whenever $|z-a| < r_0$. [The expansion in (1) on the R.H.S. is called the Taylor's series expansion of $f(z)$ in powers of $(z-a)$ or Taylor's series expansion of $f(z)$ around (or about) $z=a$.]

Proof. Consider the circle c_0 with centre at $z=a$ and radius r_0 .

Let z be any fixed point within c_0 .

Let $|z-a| = r$. Now since z is within c_0 , we have $r < r_0$. Consider a circle c_1 with centre at $z=a$ and radius r_1 where $\pi < r_1 < r_0$.

Let z' be any point on c_1 . Notice that z is within c_1 . Also $f(z)$ is analytic within c_0 and hence is analytic on and within c_1 .

Since $f(z)$ is analytic on and within c_1 and z is a point within c , using Cauchy's integral formula we have*

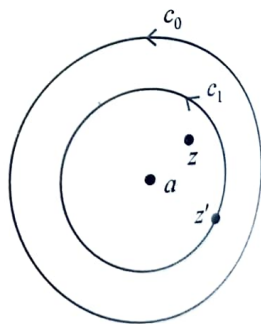


Fig. 5.1

* Cauchy's integral formula states that if $f(z)$ is analytic on and within c and $z=a$ is within c , then

$$f(a) = \frac{1}{2\pi i} \int_c \frac{f(z)}{z-a} dz.$$

In this instead of ' c ', take ' c_1 '. Instead of ' a ' take ' z '.

Instead of z in the integrand take z' ($= a$ new variable of integration).

$$f(z) = \frac{1}{2\pi i} \int_{c_1} \frac{f(z')}{z' - z} dz' \quad (\text{see footnote}) \quad \dots(2)$$

$$\text{We can observe that } \frac{1}{z' - z} = \frac{1}{(z' - a) - (z - a)} = \frac{1}{(z' - a) \left[1 - \frac{z - a}{z' - a} \right]} \quad \dots(3)$$

Also if α is any complex number other than 1,

$$\frac{1}{1 - \alpha} = 1 + \alpha + \alpha^2 + \dots + \alpha^{n-1} + \frac{\alpha^n}{1 - \alpha} \quad \dots(4)$$

[We can verify (4) simplifying directly the right hand side of (4). Or, see that (4) is true by directly multiplying (4) with $1 - \alpha$.]

Let us take $\alpha = \frac{z - a}{z' - a}$ in (3) and use (4).

$$\text{We get } \frac{1}{z' - z} = \frac{1}{z' - a} \left[1 + \alpha + \alpha^2 + \dots + \alpha^{n-1} + \frac{\alpha^n}{1 - \alpha} \right] \quad \dots(5)$$

Multiplying both sides by $f(z')/2\pi i$

$$\begin{aligned} \therefore \frac{f(z')}{2\pi i (z' - z)} &= \frac{1}{2\pi i} \left[\frac{f(z')}{z' - a} + \frac{f(z')}{(z' - a)^2} (z - a) + \frac{f(z')}{(z' - a)^3} (z - a)^2 \right. \\ &\quad \left. + \dots + \frac{f(z')}{(z' - a)^n} (z - a)^{n-1} + \frac{f(z')}{(z' - a)^n} (z' - a)^n \right] \quad \dots(6) \end{aligned}$$

Integrating both sides around the circle c_1 in the counter clockwise sense and using

$$f^n(a) = \frac{n!}{2\pi i} \int_{c_1} \frac{f(z')}{(z' - a)^{n+1}} dz' \quad \dots(7)$$

we get from (3) that

$$f(z) = f(a) + f'(a)(z - a) + \frac{f''(a)}{2!} (z - a)^2 + \dots + \frac{f^{n-1}(a)}{(n-1)!} (z - a)^{n-1} + R_n \quad \dots(8)$$

$$\text{where } R_n = \frac{(z - a)^n}{2\pi i} \int_{c_1} \frac{f(z') dz'}{(z' - z)(z' - a)^n} \quad \dots(9)$$

We have $|z - a| = r$, $|z' - a| = r_1$ and $|z' - z| \geq r_1 - r$. If M denotes the maximum value of $|f(z')|$ on c_1 (which exists) then

$$|R_n| \leq \frac{r^n}{2\pi} \cdot \frac{2\pi r_1 M}{(r_1 - r) r_1^n} \leq \frac{r_1 M}{r_1 - r} \left(\frac{r}{r_1} \right)^n$$

But $0 < r/r_1 < 1$.

Hence, as $n \rightarrow \infty$, $R_n \rightarrow 0$. Hence, allowing n to tend to zero in (8), we get

$$f(z) = f(a) + f'(a)(z - a) + \frac{f''(a)}{2!} (z - a)^2 + \frac{f'''(a)}{3!} (z - a)^3 + \dots$$

Hence, we have the following representation for $f(z)$ as a power series in powers of $(z-a)$:

$$f(z) = f(a) + \sum_{n=1}^{\infty} \frac{(z-a)^n}{n!} f^n(a) = \sum_{n=0}^{\infty} \frac{f^n(a)}{n!} (z-a)^n$$

This is Taylor's series. It represents the function $f(z)$ which is analytic at all points inside a circle having its centre at a . The largest circle which can be drawn around $z = a$, such that $f(z)$ is analytic throughout its interior, is called the circle of convergence of the Taylor's series of $f(z)$. The radius of this circle is called the radius of convergence of the series.

Special case. If we take $a = 0$ in Taylor's series expansion, we get

$$f(z) = f(0) + f'(0) \cdot z + \frac{f''(0)}{2!} z^2 + \frac{f'''(0)}{3!} z^3 + \dots = f(0) + \sum_{n=1}^{\infty} \frac{f^n(0)}{n!} z^n$$

This is called Maclaurin's series expansion of $f(z)$.

Some important observations

1. Suppose we want the Taylor's series expansion of $f(z)$ around $z = a$. Then, $f(z)$ must be analytic at $z = a$. Consider a circle $c : |z - a| = R$ where R is as large as possible such that within c , $f(z)$ is analytic everywhere.

$$\text{Then } f(z) = \sum_{n=0}^{\infty} \frac{f^n(a)}{n!} (z-a)^n$$

is the Taylor's series expansion of $f(z)$ about $z = a$. We need not verify the convergence of the series on the R.H.S. The series will be convergent to $f(z)$ for $|z - a| < R$.

The maximum R in the above is the distance from the point $z = a$ to the nearest possible singular point of $f(z)$.

2. As in the case of functions of real variables, we can derive Maclaurin's series expansions of standard functions : e^z , $\sin z$, $\cos z$, $\sinh z$, $\cosh z$, $\frac{1}{1+z}$ etc.

We have the following results which are same as those in real variable theory :

$$(a) \quad e^z = 1 + z + \frac{z^2}{2!} + \frac{z^3}{3!} + \dots + \frac{z^n}{n!} + \dots = \sum_{n=0}^{\infty} \frac{z^n}{n!} \quad \text{for all } z, \text{ i.e., } |z| < \infty$$

$$(b) \quad \sin z = z - \frac{z^3}{3!} + \frac{z^5}{5!} - \dots = \sum_{n=0}^{\infty} (-1)^n \frac{z^{(2n+1)}}{(2n+1)!} \quad \text{for } |z| < \infty$$

$$(c) \quad \cos z = 1 - \frac{z^2}{2!} + \frac{z^4}{4!} - \dots = \sum_{n=0}^{\infty} (-1)^n \frac{z^{2n}}{(2n)!} \quad \text{for } |z| < \infty$$

$$(d) \quad \sinh z = \sum_{n=0}^{\infty} \frac{z^{2n+1}}{(2n+1)!} \quad \text{for } |z| < \infty$$

$$(e) \quad \cosh z = \sum_{n=0}^{\infty} \frac{z^{2n}}{(2n)!} \quad \text{for } |z| < \infty$$

$$(f) \quad \frac{1}{1+z} = \sum_{n=0}^{\infty} (-1)^n z^n, \quad \text{for } |z| < 1$$

$$(g) \frac{1}{1-z} = \sum_{n=0}^{\infty} z^n, \text{ for } |z| < 1$$

$$(h) \frac{1}{(1-z)^m} = \sum_{n=0}^{\infty} \frac{(m+n-1)!}{(m-1)!(n)!} z^n, \text{ for } |z| < 1$$

$$(i) \frac{1}{(1+z)^m} = \sum_{n=0}^{\infty} \binom{-m}{n} z^n = 1 - mz + \frac{m(m+1)}{2!} z^2 - \frac{m(m+1)(m+2)}{3!} z^3 + \dots$$

Since e^z , $\sin z$, $\cos z$, $\sinh z$, $\cosh z$ are all entire functions, [i.e., analytic everywhere], their Maclaurin's series expansions are valid for all z .

$\frac{1}{1+z}$ has $z = -1$ as its singular point. The Maclaurin's series expansion is valid in $|z| < R$ where R is large enough such that $z = -1$ is not within the circle. Since $z = -1$ is the singular point nearest to $z = 0$, the expansion of $\frac{1}{1+z}$ is valid only for $|z| < 1$.

5.3 An Important Suggestion for Working Problems in Taylor's Series

To obtain the Taylor's series expansion of $f(z)$ around/about $z = a$, put $z - a = w$. Then $f(z) = f(w + a) = \phi(w)$ say. Write the Maclaurin's series expansion of $\phi(w)$ either by using standard expansions or directly. Then substitute $w = z - a$. We get the required Taylor's series expansion.

We present here under some examples to explain the procedure :

SOLVED EXAMPLES

Example 1 : Obtain the Taylor series expansion of $f(z) = \frac{1}{z}$ about the point $z = 1$.

Solution : [At $z = 1$, $f(z)$ is analytic.]

The point $z = 0$ is the only singular point and is at a distance of 1 unit from $z = 1$.

Hence, the Taylor's series expansion around $z = 1$ is valid in $|z - 1| < 1$.

We want the Taylor's series expansion of $f(z) = \frac{1}{z}$ about $z = 1$.

Put $z - 1 = w \quad \therefore \quad z = 1 + w$

$$\begin{aligned} \therefore f(z) &= \frac{1}{z} = \frac{1}{1+w} = (1+w)^{-1} \\ &= 1 - w + w^2 - w^3 + \dots \quad \text{for } |w| < 1 \\ &= 1 - (z-1) + (z-1)^2 - (z-1)^3 + \dots \quad \text{for } |z-1| < 1. \end{aligned}$$

This is the required expansion.

Example 2 : (i) Expand e^z as Taylor's series about $z = 1$.

[JNTU Nov. 2008 (Set No.1), JNTU (H) Nov. 2009 (Set No.3)]

(ii) Find the Taylor's series expansion of e^z about $z = 3$.

[JNTU (A) Dec. 2009 (Set No.1,4)]

Solution : (i) We want the Taylor's series expansion of e^z around $z = 1$.

Put $z - 1 = w \quad \therefore \quad z = 1 + w$

$$\therefore e^z = e^{1+w} = e \cdot e^w$$

$$= e \left[1 + w + \frac{w^2}{2!} + \frac{w^3}{3!} + \dots \right] \quad \text{for all } w$$

$$= e \left[1 + (z-1) + \frac{(z-1)^2}{2!} + \frac{(z-1)^3}{3!} + \dots \right]$$

This can also be written as $e^z = e + e \sum_{n=1}^{\infty} \frac{(z-1)^n}{n!}$ if $|z-1| < \infty$, i.e., for all z .

(ii) Proceeding as in (i), we obtain

$$e^z = e^{3+w} = e^3 \cdot e^w = e^3 \left[1 + w + \frac{w^2}{2!} + \frac{w^3}{3!} + \dots \right]$$

$$= e^3 \left[1 + (z-3) + \frac{(z-3)^2}{2!} + \frac{(z-3)^3}{3!} + \dots \right] \quad \text{if } |z-3| < \infty \text{ i.e., for all } z.$$

Example 3 : Expand $f(z) = \frac{1}{z^2}$

(a) in powers of $z + 1$ [or, Prove that when $|z + 1| < 1$, $z^{-2} = 1 + \sum_{n=1}^{\infty} (n+1)(z+1)^n$]

(b) in powers of $z - 2$. State the region of validity of the expansion in each case.

Solution : $f(z) = \frac{1}{z^2}$ has $z = 0$ as its (only) singular point.

(a) To expand in powers of $(z + 1)$:

Put $z + 1 = w \quad \therefore \quad z = w - 1$.

$$\text{Hence, } f(z) = \frac{1}{z^2} = \frac{1}{(w-1)^2} = \frac{1}{(1-w)^2}$$

[Recall that

$$\frac{1}{1-w} = 1 + w + w^2 + \dots \quad \text{if } |w| < 1$$

$$\text{and } \frac{1}{(1-w)^2} = 1 + 2w + 3w^2 + \dots \quad \text{if } |w| < 1.]$$

$$\text{Hence, } f(z) = \frac{1}{(1-w)^2} = 1 + 2w + 3w^2 + 4w^3 + \dots \quad \text{if } |w| < 1$$

$$= 1 + \sum_{n=1}^{\infty} (n+1)w^n \text{ if } |w| < 1$$

$$= 1 + \sum_{n=1}^{\infty} (n+1)|z+1|^n \text{ if } |z+1| < 1$$

Aliter . Also Refer Solved Example 23 on Page 328.

(b) To expand in powers of $z - 2$, put $z - 2 = w$

$$\therefore z = 2 + w$$

$$\therefore f(z) = \frac{1}{z^2} = \frac{1}{(2+w)^2} = \frac{1}{4\left(1+\frac{w}{2}\right)^2} = \frac{1}{4} \cdot \frac{1}{\left(1+\frac{w}{2}\right)^2}$$

$$\text{We have } \frac{1}{(1+z)^2} = 1 - 2z + 3z^2 - 4z^3 + \dots \text{ if } |z| < 1$$

Taking $z = w/2$ in the above

$$\frac{1}{1+\left(\frac{w}{2}\right)^2} = 1 - 2 \cdot \left(\frac{w}{2}\right) + 3 \cdot \left(\frac{w}{2}\right)^2 - 4 \left(\frac{w}{2}\right)^3 \dots \text{ if } \left|\frac{w}{2}\right| < 1.$$

Hence,

$$f(z) = \frac{1}{z^2} = \frac{1}{4} \left[1 - \frac{2w}{2} + 3 \left(\frac{w}{2}\right)^2 - 4 \left(\frac{w}{2}\right)^3 \dots \right] \text{ if } \left|\frac{w}{2}\right| < 1$$

$$= \frac{1}{4} + \frac{1}{4} \sum_{n=1}^{\infty} (-1)^n (n+1) \left(\frac{w}{2}\right)^n \text{ if } |w| < 2$$

$$= \frac{1}{4} + \frac{1}{4} \sum_{n=1}^{\infty} (-1)^n (n+1) \left(\frac{z-2}{2}\right)^n \text{ if } |z-2| < 2.$$

Example 4 : Expand $\sinh z$ by Taylor's series about $z = \pi i$.

Solution : We want the Taylor's series expansion of $f(z) = \sinh z$ about $z = \pi i$.

$$\text{Put } w = z - \pi i \Rightarrow z = w + \pi i$$

$$\therefore f(z) = \sinh(w + \pi i) = \sinh w \cosh \pi i + \cosh w \sinh \pi i$$

$$\text{But } \cosh \pi i = \frac{e^{\pi i} + e^{-\pi i}}{2} = \frac{2 \cos \pi}{2} = -1$$

$$\text{and } \sinh \pi i = \frac{e^{\pi i} - e^{-\pi i}}{2} = \frac{2i \sin \pi}{2} = 0.$$

$$\text{Hence } f(z) = -\sinh w$$

$$= - \left[w + \frac{w^3}{3!} + \frac{w^5}{5!} + \dots \right] \text{ for all } w.$$

$$= - \sum_{n=0}^{\infty} \frac{w^{2n+1}}{(2n+1)!} = - \sum_{n=0}^{\infty} \frac{(z-\pi i)^{2n+1}}{(2n+1)!} \text{ for all } z.$$

This is the required Taylor's series expansion of $\sinh z$ around $z = \pi i$.

Example 5 : Within what circle does the Maclaurin's series for the function $\tanh z$ converge to the function.

Solution : Let $f(z) = \tanh z = \frac{\sinh z}{\cosh z}$

The function is not analytic whenever $\cosh z = 0$

i.e., $z = \pm \frac{\pi i}{2}, \pm \frac{3\pi i}{2}$ etc.

We note that $f(z) = \tanh z$ is analytic at $z = 0$ and the singular points $\pm \pi i / 2$ are the nearest to $z = 0$ and are at a distance of $\frac{\pi}{2}$ from $z = 0$.

Hence, the Maclaurin's series expansion of $\tanh z$ will be valid for the region $|z| < \pi/2$.

Example 6 : Show that $(1+z)^{-1} = \sum_{n=0}^{\infty} (-1)^n z^{-n-1}$ for $|z| > 1$.

Solution : $f(z) = (1+z)^{-1} = \frac{1}{(1+z)} = \frac{1}{z\left(1+\frac{1}{z}\right)} = \frac{1}{z}\left(1+\frac{1}{z}\right)^{-1}$

$$= \frac{1}{z} \left[1 - \frac{1}{z} + \frac{1}{z^2} - \frac{1}{z^3} + \dots + \dots \right], \text{ if } \left| \frac{1}{z} \right| < 1 \text{ and } z \neq 0$$

$$= \frac{1}{z} - \frac{1}{z^2} + \frac{1}{z^3} - \frac{1}{z^4} + \dots + \dots, \text{ if } |z| > 1$$

$$= \sum_{n=0}^{\infty} (-1)^n z^{-n-1}, \text{ if } |z| > 1.$$

Example 7 : Expand $f(z) = \frac{z-1}{z+1}$ in Taylor's series about the point

(i) $z = 0$ and (ii) $z = 1$.

[JNTU Nov. 2008 (Set No.4)]

Solution : (i) $f(z) = \frac{z-1}{z+1} = \frac{z+1-2}{z+1} = 1 - \frac{2}{z+1} = 1 - 2 \cdot \frac{1}{(1+z)} = 1 - 2(1+z)^{-1}$

$$= 1 - 2 \left(1 - z + z^2 - z^3 + \dots + \dots \right) \text{ if } |z| < 1$$

$$= -1 + 2 \left(z - z^2 + z^3 - \dots + \dots \right) \text{ if } |z| < 1$$

$$= -1 + 2 \sum_{n=1}^{\infty} (-1)^{n-1} z^n \text{ if } |z| < 1$$

(ii) To expand $f(z)$ about $z = 1$. Put $z - 1 = w \therefore z = 1 + w$.

Hence,

$$\begin{aligned}
 f(z) &= \frac{z-1}{z+1} = \frac{w}{1+w+1} = \frac{w}{2+w} = \frac{w}{2\left(1+\frac{w}{2}\right)} = \frac{w}{2} \left(1+\frac{w}{2}\right)^{-1} \\
 &= \frac{w}{2} \left[1 - \frac{w}{2} + \left(\frac{w}{2}\right)^2 - \left(\frac{w}{2}\right)^3 + \dots \right] \text{ if } \left| \frac{w}{2} \right| < 1 \\
 &= \frac{w}{2} - \left(\frac{w}{2}\right)^2 + \left(\frac{w}{2}\right)^3 - \left(\frac{w}{2}\right)^4 + \dots + \dots \text{ if } |w| < 2 \\
 &= \sum_{n=1}^{\infty} (-1)^{n-1} \left(\frac{z-1}{2}\right)^n \text{ if } |z-1| < 2.
 \end{aligned}$$

Example 8 : Expand $f(z) = \frac{z-1}{z^2}$ in a Taylor's series in powers of $z-1$ and determine the region of convergence.

Solution : Put $z-1 = w$. Then $z = 1+w$

$$\begin{aligned}
 \therefore f(z) &= \frac{w}{(1+w)^2} = w(1+w)^{-2} \\
 &= w \left[1 - 2w + 3w^2 - 4w^3 + \dots \right] \text{ if } |w| < 1 \\
 &= w - 2w^2 + 3w^3 - 4w^4 + \dots + \dots \text{ if } |w| < 1 \\
 &= \sum_{n=0}^{\infty} (-1)^{n+1} (n+1) w^n \text{ if } |w| < 1 \\
 &= \sum_{n=0}^{\infty} (-1)^{n+1} (n+1) (z-1)^n \text{ if } |z-1| < 1.
 \end{aligned}$$

An Alternative Method.

$$\text{Let, } f(z) = \frac{z-1}{z^2} = \frac{1}{z} - \frac{1}{z^2} \quad \dots (1)$$

$$\text{By Taylor's theorem, } f(z) = f(a) + \sum_{n=1}^{\infty} \frac{(z-a)^n}{n!} f^n(a)$$

$$\text{Taking } a = 1, \text{ we get } f(z) = f(1) + \sum_{n=1}^{\infty} \frac{(z-1)^n}{n!} f^n(1) \quad \dots (2)$$

$$\text{Differentiating (1) } n \text{ times, we get } f^n(z) = \frac{(-1)^n n!}{z^{n+1}} + \frac{(-1)^{n+1} (n+1)!}{z^{n+2}}$$

$$\begin{aligned}\therefore f^n(1) &= (-1)^n n! + (-1)^{n+1} (n+1)! \\ &= (-1)^n n! [1 + (-1)(n+1)] = (-1)^n \cdot n! \cdot (-n) = (-1)^{n+1} \cdot n \cdot n!\end{aligned}$$

and $f(1) = 1 - 1 = 0$

Hence,
$$f(z) = 0 + \sum_{n=1}^{\infty} \frac{(z-1)^n}{n!} (-1)^{n+1} \cdot n \cdot n! \quad [\text{by (2)}]$$

$$= \sum_{n=1}^{\infty} (-1)^{n+1} n (z-1)^n$$

which is the required Taylor's series.

The region of convergence is $|z - 1| < 1$. We note that $z = 0$ is a singular point of $f(z)$. We want to expand $f(z)$ about $z = 1$. The nearest singular point from $z = 1$ is $z = 0$.

Hence, if $|z - 1| < R = 1$, i.e., $|z - 1| < R$, the expansion converges to $f(z)$. Hence, the region of convergence is $\{z \mid |z - 1| < 1\}$.

Example 9 : Find Taylor's expansion for the function $f(z) = \frac{1}{(1+z)^2}$ with centre at $-i$.
[JNTU 1998S, JNRU (A) Dec. 2009 (Set No. 3)]

Solution : By Taylor's theorem

$$f(z) = f(a) + (z-a)f'(a) + \frac{(z-a)^2}{2!} f''(a) + \dots + \frac{(z-a)^n}{n!} f^n(a) + \dots$$

Put $a = -i$.

$$\therefore f(z) = f(-i) + (z+i) f'(-i) + \frac{(z+i)^2}{2!} f''(-i) + \dots + \frac{(z+i)^n}{n!} f^n(-i) + \dots \quad \dots (1)$$

Here $f(z) = \frac{1}{(1+z)^2}$

$$\therefore f^n(z) = (-1)^n \frac{(n+1)!}{(1+z)^{n+2}}$$

$$\therefore f(-i) = \frac{1}{(1-i)^2} = \frac{i}{2} \quad \text{and} \quad f^n(-i) = \frac{(-1)^n (n+1)!}{(1-i)^{n+2}}$$

Substituting in (1), we get

$$\begin{aligned}\frac{1}{(1+z)^2} &= \frac{i}{2} + \sum_{n=1}^{\infty} \frac{(z+i)^n}{n!} \frac{(-1)^n (n+1)!}{(1-i)^{n+2}} = \frac{i}{2} + \sum_{n=1}^{\infty} (-1)^n (n+1) \frac{(z+i)^n}{(1-i)^{n+2}} \\ &= \frac{i}{2} + \sum_{n=1}^{\infty} (-1)^n (n+1) \frac{(z+i)^n}{(1-i)^n} \frac{1}{(1-i)^2} = \frac{i}{2} + \sum_{n=1}^{\infty} (-1)^n (n+1) \frac{(z+i)^n}{(1-i)^n (-2i)} \\ &= \frac{i}{2} + \frac{i}{2} \sum_{n=1}^{\infty} (-1)^n (n+1) \frac{(z+i)^n}{(1-i)^n}\end{aligned}$$

Example 10 : Obtain the Taylor expansion of $e^{(1+z)}$ in the powers of $(z-1)$.

[JNTU 2003, 2004 (Set No. 4), JNTU (A) Dec. 2009 (Set No. 3)]

Solution : Direct method.

By Taylor's theorem
$$f(z) = \sum_{n=0}^{\infty} (z-a)^n \cdot \frac{f^n(a)}{n!}$$

Taking $a = 1$ and $f(z) = e^{1+z}$, we get
$$e^{1+z} = \sum_{n=0}^{\infty} (z-1)^n \frac{e^{1+1}}{n!} \quad [\because f^n(z) = e^{1+z}]$$

$$= e^2 \sum_{n=0}^{\infty} \frac{(z-1)^n}{n!}$$
 which is the required Taylor's series.

Indirect method.
$$e^{1+z} = e^{2+(z-1)} = e^2 \cdot e^{z-1} = e^2 \sum_{n=0}^{\infty} \frac{(z-1)^n}{n!} \quad \left[\because e^z = \sum_{n=0}^{\infty} \frac{z^n}{n!} \right]$$

Example 11 : Find the Taylor's series expansion of $\frac{1}{z^2+z-6}$ about $z = -1$.

Solution : Let
$$f(z) = \frac{1}{z^2+z-6} = \frac{1}{(z+3)(z-2)}$$
$$= \frac{1}{5} \left[\frac{1}{z-2} - \frac{1}{z+3} \right] = \frac{1}{5} \left[\frac{1}{z+1-3} - \frac{1}{z+1+2} \right]$$
$$= \frac{1}{5} \cdot \left(\frac{-1}{3} \right) \left[\frac{1}{1 - \left(\frac{z+1}{3} \right)} \right] - \frac{1}{5} \cdot \frac{1}{2} \left[\frac{1}{1 + \frac{z+1}{2}} \right]$$

Expanding by binomial series

$$f(z) = -\frac{1}{15} \sum_{n=0}^{\infty} \left(\frac{z+1}{3} \right)^n - \frac{1}{10} \sum_{n=0}^{\infty} (-1)^n \left(\frac{z+1}{2} \right)^n$$

valid when $\left| \frac{z+1}{3} \right| < 1$ and $\left| \frac{z+1}{2} \right| < 1$.

Example 12 : Obtain the Taylor's series to represent the function $\frac{z^2-1}{(z+2)(z+3)}$ in the region $|z| < 2$.

[JNTU 2005S, April 2006, JNTU (H) Nov. 2009 (Set No. 4)]

Solution : Let $f(z) = \frac{z^2-1}{(z+2)(z+3)} = 1 + \frac{3}{z+2} - \frac{8}{z+3}$ (Resolving into partial fractions)

$$= 1 + \frac{3}{2 \left(1 + \frac{z}{2} \right)} - \frac{8}{3 \left(1 + \frac{z}{3} \right)} = 1 + \frac{3}{2} \left(1 + \frac{z}{2} \right)^{-1} - \frac{8}{3} \left(1 + \frac{z}{3} \right)^{-1}$$

Expanding by binomial series

$$\begin{aligned} f(z) &= 1 + \frac{3}{2} \left(1 - \frac{z}{2} + \frac{z^2}{4} - \frac{z^3}{8} + \dots \right) - \frac{8}{3} \left(1 - \frac{z}{3} + \frac{z^2}{9} - \frac{z^3}{27} + \dots \right) \\ &= 1 + \frac{3}{2} \sum_{n=0}^{\infty} \frac{(-1)^n}{2^n} \cdot z^n - \frac{8}{3} \sum_{n=0}^{\infty} \frac{(-1)^n}{3^n} z^n = 1 + \sum_{n=0}^{\infty} (-1)^n \left[\frac{3}{2^{n+1}} - \frac{8}{3^{n+1}} \right] z^n \end{aligned}$$

which is the required Taylor's series.

Example 13 : Find the Taylor's series expansion of $\cosh z$ about $z = \pi i$

[JNTU Nov. 2006 (Set No. 3)]

Solution : Let $f(z) = \cosh z$

Put $w = z - \pi i$. Then $z = w + \pi i$

$$\therefore f(z) = \cosh(w + \pi i) = \cosh w \cosh \pi i - \sinh w \sinh \pi i$$

$$\text{But } \cosh \pi i = \cos \pi \left[(\text{or}) \frac{e^{\pi i} + e^{-\pi i}}{2} = \frac{2 \cos \pi}{2} \right] = -1$$

$$\text{and } \sinh \pi i = i \sin \pi \left[(\text{or}) \frac{e^{\pi i} - e^{-\pi i}}{2} = \frac{2i \sin \pi}{2} \right] = 0$$

$$\therefore f(z) = -\cosh w = - \left[1 + \frac{w^2}{2!} + \frac{w^4}{4!} + \dots \right] \text{ for all } w$$

$$= - \sum_{n=0}^{\infty} \frac{w^{2n}}{2n!} = - \sum_{n=0}^{\infty} \frac{(z - \pi i)^{2n}}{2n!}$$

which is the required Taylor's series.

An Alternate method.

$$\begin{aligned} f(z) = \cosh z &= \frac{e^z + e^{-z}}{2} = \frac{e^{z - \pi i + \pi i} + e^{-z + \pi i - \pi i}}{2} \\ &= \frac{1}{2} e^{\pi i} \cdot e^{z - \pi i} + \frac{e^{-\pi i}}{2} e^{-(z - \pi i)} \\ &= -\frac{1}{2} \sum_{n=0}^{\infty} \frac{(z - \pi i)^n}{n!} - \frac{1}{2} \sum_{n=0}^{\infty} \frac{(-1)^n (z - \pi i)^n}{n!} = - \sum_{n=0}^{\infty} \frac{(z - \pi i)^{2n}}{(2n)!} \end{aligned}$$

Example 14 : Expand $\log(1 - z)$ when $|z| < 1$ using Taylor series.

[JNTU 2003S, Nov. 2008 (Set No. 3)]

Solution : Let $f(z) = \log(1 - z)$ $\therefore f(0) = \log 1 = 0$

$$\text{Also } f'(z) = \frac{-1}{(1-z)} \quad \therefore f'(0) = -1$$

$$f''(z) = \frac{-1}{(1-z)^2} \quad \therefore f''(0) = -1$$

$$f'''(z) = \frac{-2}{(1-z)^3} \quad \therefore f'''(0) = -2 \text{ and so on.}$$

By Taylor's theorem about $z = 0$ is

$$f(z) = f(0) + z f'(0) + \frac{z^2}{2!} f''(0) + \frac{z^3}{3!} f'''(0) + \dots$$

$$= 0 - z - \frac{z^2}{2!} - \frac{2z^3}{3!} + \dots = - \left(z + \frac{z^2}{2} + \frac{z^3}{3} + \dots \right)$$

which is the required Taylor's series.

Example 15 : Find Taylor's expansion of $f(z) = \frac{2z^3+1}{z^2+z}$ about the point

(i) $z = i$ [JNTU 1998] (ii) $z = 1$.

[JNTU April, Nov. 2006 (Set No. 2)]

Solution : Given $f(z) = \frac{2z^3+1}{z^2+z} = 2z - 2 + \frac{2z+1}{z(z+1)}$ (Resolving into partial fractions)

$$= 2(z-1) + \frac{2}{z+1} + \frac{1}{z(z+1)} = 2(z-1) + \frac{2}{z+1} + \frac{1}{z} - \frac{1}{z+1}$$

$$= 2(z-1) + \frac{1}{z} + \frac{1}{z+1} \quad \dots (1)$$

Differentiating (1) 'n' times,

$$f^n(z) = (-1)^n n! \left[\frac{1}{z^{n+1}} + \frac{1}{(z+1)^{n+1}} \right] \quad \dots (2)$$

$$\therefore f^n(i) = (-1)^n n! \left[\frac{1}{(i)^{n+1}} + \frac{1}{(i+1)^{n+1}} \right] \quad \dots (3)$$

$$\text{By Taylor's theorem } f(z) = f(a) + \sum_{n=1}^{\infty} \frac{(z-a)^n}{n!} f^n(a) \quad \dots (4)$$

(i) To find the Taylor's series about $z = i$

Putting $a = i$, we get

$$\begin{aligned} f(z) &= f(i) + \sum_{n=1}^{\infty} \frac{(z-i)^n}{n!} f^n(i) \\ &= \frac{i}{2} - \frac{3}{2} + \sum_{n=1}^{\infty} (-1)^n (z-i)^n \left[\frac{1}{i^{n+1}} + \frac{1}{(i+1)^{n+1}} \right], \text{ using (3)} \\ &= \frac{i}{2} - \frac{3}{2} + (z-i) \left(3 + \frac{i}{z} \right) + \dots \end{aligned}$$

(ii) To find the Taylor's series about $z = 1$

Singularities of $f(z)$ are given by $z = 0$ and $z = -1$. Draw a circle with centre at $z = 1$ and radius 1. Then within the circle $|z - 1| = 1$, the function $f(z)$ is analytic. Thus $f(z)$ can be expanded in a Taylor's series within the circle $|z - 1| = 1$, which is the circle of convergence.

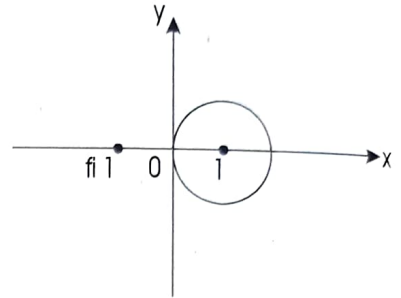
From (2), we have

$$f^n(z) = (-1)^n n! \left[\frac{1}{z^{n+1}} + \frac{1}{(z+1)^{n+1}} \right]$$

$$\therefore f^n(1) = (-1)^n n! \left(\frac{1}{1^{n+1}} + \frac{1}{2^{n+1}} \right) = (-1)^n n! \left(1 + \frac{1}{2^{n+1}} \right)$$

By Taylor's theorem,

$$\begin{aligned} f(z) &= f(1) + \sum_{n=1}^{\infty} \frac{(z-1)^n}{n!} f^n(1) \text{ [using (4)]} \\ &= \frac{3}{2} + \sum_{n=1}^{\infty} \frac{(z-1)^n}{n!} \cdot (-1)^n n! \left[1 + \frac{1}{2^{n+1}} \right] = \frac{3}{2} + \sum_{n=1}^{\infty} (-1)^n \left(1 + \frac{1}{2^{n+1}} \right) (z-1)^n \end{aligned}$$



Example 16 : Expand $f(z) = \sin z$ in Taylor's series about

(i) $z = \frac{\pi}{4}$ [JNTU 2005S (Set No. 3), June 2009S (Set No. 2)]

(ii) $z = \frac{\pi}{2}$ [JNTU (A) Dec. 2009 (Set No. 1, 4)]

Solution : By Taylor's theorem,

$$f(z) = f(a) + (z-a)f'(a) + \frac{(z-a)^2}{2!} f''(a) + \dots + \frac{(z-a)^n}{n!} f^n(a) + \dots \quad \dots (1)$$

(i) Put $a = \frac{\pi}{4}$ in (1).

$$\text{Then } f(z) = f\left(\frac{\pi}{4}\right) + \left(z - \frac{\pi}{4}\right) f'\left(\frac{\pi}{4}\right) + \frac{\left(z - \frac{\pi}{4}\right)^2}{2!} f''\left(\frac{\pi}{4}\right) + \dots \quad \dots (2)$$

Now $f(z) = \sin z$

$$f'(z) = \cos z$$

$$f''(z) = -\sin z$$

$$f'''(z) = -\cos z$$

and so on.

$$f\left(\frac{\pi}{4}\right) = \frac{1}{\sqrt{2}}$$

$$f'\left(\frac{\pi}{4}\right) = \frac{1}{\sqrt{2}}$$

$$f''\left(\frac{\pi}{4}\right) = -\frac{1}{\sqrt{2}}$$

$$f'''\left(\frac{\pi}{4}\right) = -\frac{1}{\sqrt{2}}$$

Substituting in (2), we get

$$f(z) = \frac{1}{\sqrt{2}} + \left(z - \frac{\pi}{4}\right) \frac{1}{\sqrt{2}} + \frac{\left(z - \frac{\pi}{4}\right)^2}{2!} \left(-\frac{1}{\sqrt{2}}\right) + \frac{\left(z - \frac{\pi}{4}\right)^3}{3!} \left(-\frac{1}{\sqrt{2}}\right) + \dots$$

$$\text{i.e. } \sin z = \frac{1}{\sqrt{2}} \left[1 + \left(z - \frac{\pi}{4}\right) - \frac{1}{2!} \left(z - \frac{\pi}{4}\right)^2 - \frac{1}{3!} \left(z - \frac{\pi}{4}\right)^3 + \dots \right]$$

(ii) This is left as an exercise to the student.

Example 17 : Obtain the Taylor series expansion of $f(z) = \frac{e^z}{z(z+1)}$ about $z = 2$.

[JNTU 2005 S(Set No. 1, 2), JNTU (H) Nov. 2009 (Set No. 2)]

Solution : Given $f(z) = \frac{e^z}{z(z+1)} = \frac{e^{z-2} \cdot e^2}{(z-2+2)(z-2+3)}$

$$= e^2 \frac{e^{z-2}}{2 \left[1 + \frac{z-2}{2} \right]} \cdot \frac{1}{3 \left[1 + \frac{z-2}{3} \right]}$$

$$= \frac{e^2}{6} \cdot e^{z-2} \cdot \left(1 + \frac{z-2}{2} \right)^{-1} \left(1 + \frac{z-2}{3} \right)^{-1} \text{ valid when } \left| \frac{z-2}{2} \right| < 1 \text{ and } \left| \frac{z-2}{3} \right| < 1$$

$$= \frac{e^2}{6} \left[\sum_{n=0}^{\infty} \frac{(z-2)^n}{n!} \right] \left[\sum_{n=0}^{\infty} (-1)^n \left(\frac{z-2}{2} \right)^n \right] \left[\sum_{n=0}^{\infty} (-1)^n \left(\frac{z-2}{3} \right)^n \right]$$

$$= \frac{e^2}{6} \left[1 + \frac{z-2}{1!} + \frac{(z-2)^2}{2!} + \frac{(z-2)^3}{3!} + \dots \right] \left[1 - \frac{z-2}{2} + \frac{(z-2)^2}{4} - \frac{(z-2)^3}{8} + \dots \right]$$

$$\left[1 - \frac{z-2}{3} + \frac{(z-2)^2}{9} - \frac{(z-2)^3}{27} + \dots \right]$$

$$= \frac{e^2}{6} \left[1 - \frac{z-2}{2} + \frac{(z-2)^2}{4} + \frac{z-2}{1} - \frac{(z-2)^2}{2} + \frac{(z-2)^2}{2} + \dots \right] \left[1 - \frac{z-2}{3} + \frac{(z-2)^2}{9} - \dots \right]$$

$$= \frac{e^2}{6} \left[1 + \frac{z-2}{z} + \frac{(z-2)^2}{4} + \dots \right] \left[1 - \frac{z-2}{3} + \frac{(z-2)^2}{9} - \dots \right]$$

$$= \frac{e^2}{6} \left[1 - \frac{z-2}{3} + \frac{(z-2)^2}{9} + \frac{z-2}{2} - \frac{(z-2)^2}{6} + \frac{(z-2)^2}{4} - \frac{(z-2)^3}{12} + \dots \right]$$

$$= \frac{e^2}{6} \left[1 + \frac{z-2}{6} + \frac{7}{36} (z-2)^2 + \dots \right]$$

Example 18 : Obtain the Taylor series expansion of $f(z) = \frac{1}{(z-1)(z-2)}$ about $z = 0$.

Solution : By Partial fractions,

$$\begin{aligned} f(z) &= \frac{1}{(z-1)(z-2)} = \frac{1}{z-2} - \frac{1}{z-1} = \frac{1}{-2\left(1-\frac{z}{2}\right)} - \frac{1}{-(1-z)} = -\frac{1}{2}\left(1-\frac{z}{2}\right)^{-1} + (1-z)^{-1} \\ &= \frac{-1}{2}\left(1+\frac{z}{2}+\frac{z^2}{4}+\frac{z^3}{8}+\dots\right) + (1+z+z^2+z^3+\dots) \text{ valid when } \left|\frac{z}{2}\right| < 1 \text{ and } |z| < 1 \\ &= \frac{1}{2} + \frac{3}{4}z + \frac{7}{8}z^2 + \frac{15}{6}z^3 + \dots, |z| < 1 \end{aligned}$$

Example 19 : Expand $f(z) = \frac{1}{z^2 - z - 6}$ about (i) $z = -1$ (ii) $z = 1$

[JNTU 2004S, 2006, JNTU (H) Nov. 2009 (Set No.4)]

Solution : We have $f(z) = \frac{1}{z^2 - z - 6} = \frac{1}{(z-3)(z+2)}$

$$= \frac{1}{5} \left[\frac{1}{z-3} - \frac{1}{z+2} \right] \text{ (Resolving into partial fractions)}$$

(i) We have to expand $f(z)$ about $z = -1$ i.e. in powers of $(z+1)$.

$$\begin{aligned} \therefore f(z) &= \frac{1}{5} \left[\frac{1}{(z+1)-4} - \frac{1}{(z+1)+1} \right] \\ &= \frac{1}{5} \cdot \frac{-1}{4} \left[1 - \frac{z+1}{4} \right]^{-1} - \frac{1}{5} \cdot [1 + (z+1)]^{-1} \\ &= -\frac{1}{20} \left[1 + \left(\frac{z+1}{4} \right) + \left(\frac{z+1}{4} \right)^2 + \left(\frac{z+1}{4} \right)^3 + \dots \right] \\ &\quad - \frac{1}{5} [1 - (z+1) + (z+1)^2 - (z+1)^3 + \dots], \text{ by binomial series} \end{aligned}$$

$$\text{i.e. } f(z) = \frac{-1}{20} \sum_{n=0}^{\infty} \left(\frac{z+1}{4} \right)^n - \frac{1}{5} \sum_{n=0}^{\infty} (-1)^n (z+1)^n$$

This is valid in $\left| \frac{z+1}{4} \right| < 1$ and $|z+1| < 1$

$\therefore$ Region of convergence is $|z+1| < 1$ (common region)

(ii) We have to expand $f(z)$ about $z = 1$ i.e. in powers $(z-1)$.

$$\therefore f(z) = \frac{1}{5} \left[\frac{1}{(z-1)-2} - \frac{1}{(z-1)+3} \right]$$

$$\begin{aligned}
&= \frac{1}{5} \cdot \frac{-1}{2} \left[1 - \frac{z-1}{2} \right]^{-1} - \frac{1}{5} \cdot \frac{1}{3} \left[1 + \frac{z-1}{3} \right]^{-1} \\
&= -\frac{1}{10} \left[1 + \left(\frac{z-1}{2} \right) + \left(\frac{z-1}{2} \right)^2 + \left(\frac{z-1}{2} \right)^3 + \dots \right] \\
&\quad - \frac{1}{15} \left[1 - \left(\frac{z-1}{3} \right) + \left(\frac{z-1}{3} \right)^2 - \left(\frac{z-1}{3} \right)^3 + \dots \right]
\end{aligned}$$

$$\text{i.e. } f(z) = -\frac{1}{10} \sum_{n=0}^{\infty} \left(\frac{z-1}{2} \right)^n - \frac{1}{15} \sum_{n=0}^{\infty} (-1)^n \left(\frac{z-1}{3} \right)^n$$

$$\text{and this is valid in } \left| \frac{z-1}{2} \right| < 1 \text{ and } \left| \frac{z-1}{3} \right| < 1$$

i.e. when $|z-1| < 2$ i.e. region of convergence is the circle with centre at $z = 1$ and radius 2 (common region).

Example 20 : Obtain the expansion of $\frac{1}{(z-1)(z-3)}$ in a Taylor's series in powers of $(z-4)$ and determine the region of convergence.

Solution : Let $f(z) = \frac{1}{(z-1)(z-3)}$. Then $f(z)$ is not analytic at $z = 0$ and $z = 3$. If the centre of the circle is at $z = 4$, then the distances of the singularities $z = 1$ and $z = 3$ from the centre are 3 and 1. Hence if a circle is drawn with centre at $z = 4$ and radius 1, then within the circle $|z-4| = 1$ the given function $f(z)$ is analytic, hence it can be expanded in a Taylor's series within the circle $|z-4| = 1$, which is therefore the circle of convergence.

By partial fractions.

$$\begin{aligned}
f(z) &= \frac{1}{(z-1)(z-3)} = \frac{1}{2} \left(\frac{1}{z-3} - \frac{1}{z-1} \right) = \frac{1}{2} \left[\frac{1}{(z-4)+1} - \frac{1}{(z-4)+3} \right] \\
&= \frac{1}{2} [1 + (z-4)]^{-1} - \frac{1}{6} \left[1 + \left(\frac{z-4}{3} \right) \right]^{-1} \\
&= \frac{1}{2} \left[1 - (z-4) + (z-4)^2 - (z-4)^3 + \dots \right] - \frac{1}{6} \left[1 - \frac{z-4}{3} + \frac{(z-4)^2}{9} - \frac{(z-4)^3}{27} + \dots \right] \\
&= \left(\frac{1}{2} - \frac{1}{6} \right) + \left(-\frac{1}{2} + \frac{1}{18} \right) (z-4) + \left(\frac{1}{2} - \frac{1}{54} \right) (z-4)^2 + \dots \\
&= \frac{1}{3} - \frac{4}{9} (z-4) + \frac{13}{27} (z-4)^2 - \dots
\end{aligned}$$

Example 21 : Find the Taylor's series expansion of $f(z) = \frac{z-1}{z^2}$ about the point $z = 1$. Find its region of convergence.

Solution : The point $z = 0$ is the only singular point and is at a distance of 1 unit from $z = 1$. Hence, the Taylor's series expansion around $z = 1$ is valid in $|z - 1| < 1$, which is therefore the circle of convergence.

$$\begin{aligned} f(z) &= \frac{z-1}{z^2} = \frac{z-1}{[(z-1)+1]^2} = (z-1) [1 + (z-1)]^{-2} \\ &= (z-1) [1 - 2(z-1) + 3(z-1)^2 - 4(z-1)^3 + \dots] \text{ (Expanding by binomial series)} \\ &= (z-1) - 2(z-1)^2 + 3(z-1)^3 - 4(z-1)^4 + \dots \end{aligned}$$

This is Taylor's expansion of $f(z)$.

Example 22 : Expand in Taylor's series $\frac{z}{z^4+9}$ about the point $z = 0$.

$$\begin{aligned} \text{Solution : Let } f(z) &= \frac{z}{z^4+9} = \frac{z}{9\left(1+\frac{z^4}{9}\right)} = \frac{z}{9} \left(1+\frac{z^4}{9}\right)^{-1} \\ &= \frac{z}{9} \sum_{n=0}^{\infty} (-1)^n \left(\frac{z^4}{9}\right)^n \left[\because \frac{1}{1+x} = 1 - x + x^2 - x^3 + \dots = \sum_{n=0}^{\infty} (-1)^n x^n \right] \\ &= \frac{z}{9} \sum_{n=0}^{\infty} (-1)^n \frac{z^{4n}}{3^{2n}} = \sum_{n=0}^{\infty} (-1)^n \frac{z^{4n+1}}{3^{2n+2}} \end{aligned}$$

Example 23 : Show that when $|z+1| < 1$, $z^{-2} = 1 + \sum_{n=1}^{\infty} (n+1) (z+1)^n$

[JNTU 2004S, 2006 (Set No. 4)]

Solution : By Taylor's series expansion, we have

$$f(z) = f(a) + (z-a)f'(a) + \frac{(z-a)^2}{2!} f''(a) + \frac{(z-a)^3}{3!} f'''(a) + \dots$$

$$\text{Take } a = -1 \text{ and } f(z) = \frac{1}{z^2}$$

$$\therefore f(z) = f(-1) + (z+1)f'(-1) + \frac{(z+1)^2}{2!} f''(-1) + \frac{(z+1)^3}{3!} f'''(-1) + \dots \quad \dots(1)$$

$$\text{Now } f'(z) = \frac{-2}{z^3}, f''(z) = \frac{(-2)(-3)}{z^4}, f'''(z) = \frac{(-2)(-3)(-4)}{z^5}, \dots$$

$$\therefore f(-1) = 1, f'(-1) = 2, f''(-1) = 2.3, f'''(-1) = 2.3.4$$

On substituting these values in (1), we get

$$\begin{aligned} \frac{1}{z^2} &= 1 + (z+1).2 + \frac{(z+1)^2}{2!} (2)(3) + \frac{(z+1)^3}{3!} (2)(3)(4) + \dots \\ &= 1 + 2(z+1) + 3(z+1)^2 + 4(z+1)^3 + \dots \end{aligned}$$

$$\text{i.e., } z^{-2} = 1 + \sum_{n=1}^{\infty} (n+1)(z+1)^n$$

Region of convergence is $|z+1| < 1$.

Example 24 : Find Taylor's expansion of $f(z) = \frac{z+1}{(z-3)(z-4)}$ about the point $z = 2$. Determine the region of convergence.

Solution : If the centre of a circle is at $z = 2$, then the distances of the singularities $z = 3$ and $z = 4$ from the centre are 1 and 2. Hence if a circle is drawn with centre $z = 2$ and radius 1, then within the circle $|z - 2| = 1$, the given function $f(z)$ is analytic, hence it can be expanded in a Taylor's series within the circle $|z - 2| = 1$, which is therefore the circle of convergence.

By partial fractions,

$$\begin{aligned} f(z) &= \frac{z+1}{(z-3)(z-4)} = \frac{-4}{z-3} + \frac{5}{z-4} \\ &= \frac{-4}{(z-2)-1} + \frac{5}{(z-2)-2} = 4[1-(z-2)]^{-1} - \frac{5}{2} \left[1 - \frac{z-2}{2} \right]^{-1} \\ &= 4 \left[1 + (z-2) + (z-2)^2 + (z-2)^3 + \dots \right] - \frac{5}{2} \left[1 + \frac{z-2}{2} + \left(\frac{z-2}{2} \right)^2 + \left(\frac{z-2}{2} \right)^3 + \dots \right] \\ &= \left(4 - \frac{5}{2} \right) + \left(4 - \frac{5}{4} \right)(z-2) + \left(4 - \frac{5}{8} \right)(z-2)^2 + \left(4 - \frac{5}{16} \right)(z-2)^3 + \dots \\ &= \frac{3}{2} + \frac{11}{4}(z-2) + \frac{27}{8}(z-2)^2 + \frac{59}{16}(z-2)^3 + \dots \end{aligned}$$

This is Taylor's expansion of $f(z)$.

Example 25 : Expand $f(z) = \frac{1}{(z+1)^2}$ in Taylor's series about $z = -i$.

Solution : We want the Taylor's series expansion of $f(z)$ in powers of $z + i$.

Put $z + i = w \therefore z = w - i$

$$\therefore f(z) = \frac{1}{(z+1)^2} = \frac{1}{(w-i+1)^2} = \frac{1}{[1-i+w]^2} = \frac{1}{(1-i)^2 \left[1 + \frac{w}{1-i} \right]^2}$$

Expanding by binomial expansion,

$$\begin{aligned} f(z) &= \frac{1}{(1-i)^2} \left[1 + \frac{w}{1-i} \right]^{-2} = \frac{-1}{2i} \left[1 - \frac{2w}{1-i} + \frac{3w^2}{(1-i)^2} - \frac{4w^3}{(1-i)^3} + \dots \right] \\ &= \frac{i}{2} \left[1 - \frac{2}{1-i} (z+i) + \frac{3}{(1-i)^2} (z+i)^2 - \frac{4}{(1-i)^3} (z+i)^3 + \dots \right] \end{aligned}$$

This is the required expansion.

Example 26 : Expand $\log z$ by Taylor's series about $z = 1$.

[JNTU 2007S, 2008S, June 2009S (Set No.4), JNTU (K) Nov. 2009 (Set No. 1)]

Solution : Let $f(z) = \log z$

Then Taylor's series for $f(z)$ in powers of $z - a$ will be

$$f(z) = f(a) + (z-a)f'(a) + \frac{(z-a)^2}{2!} f''(a) + \dots$$

Taking $a = 1$, we have

$$f(z) = f(1) + (z-1)f'(1) + \frac{(z-1)^2}{2!} f''(1) + \dots \quad \dots(1)$$

$$= f(1) + \sum_{n=1}^{\infty} \frac{(z-1)^n}{n!} f^n(1)$$

Now $f(1) = \log 1 = 0$. Differentiating $f(z)$ successively, we get

$$f'(z) = \frac{1}{z}, f''(z) = -\frac{1}{z^2}, f'''(z) = \frac{2}{z^3}, f''''(z) = \frac{(-2)(3)}{z^4}, \dots$$

$$\therefore f'(1) = 1, f''(1) = -1, f'''(1) = 2, f''''(1) = -6, \dots$$

Substituting these values in (1), we get

$$\begin{aligned} f(z) &= 0 + (z-1) - \frac{(z-1)^2}{2!} + 2 \frac{(z-1)^3}{3!} - 6 \frac{(z-1)^4}{4!} + \dots \\ &= (z-1) - \frac{(z-1)^2}{2} + \frac{(z-1)^3}{3} - \frac{(z-1)^4}{4} + \dots, \text{ valid for } |z-1| < 1. \end{aligned}$$

An Alternative method :

We want the Taylor's series expansion of $\log z$ around $z = 1$.

Put $z - 1 = w \quad \therefore z = 1 + w$

$$\therefore \log z = \log(1 + w)$$

$$= w - \frac{w^2}{2} + \frac{w^3}{3} - \dots + (-1)^{n-1} \frac{w^n}{n} + \dots, \text{ for } |w| < 1$$

$$= (z-1) - \frac{1}{2}(z-1)^2 + \frac{1}{3}(z-1)^3 - \dots + (-1)^{n-1} \frac{1}{n}(z-1)^n + \dots, \text{ for } |z-1| < 1.$$

Example 27 : Find the Taylor series for $\frac{z}{z+2}$ about $z = 1$. Also find the region of convergence.

[JNTU Feb. 2008S (Set No.3)]

Solution : Let $f(z) = \frac{z}{z+2}$.

Then the singularities of $f(z)$ are given by $z + 2 = 0$ i.e., $z = -2$.

If the centre of the circle is at $z = 1$, then the distance of the singularity $z = -2$ from the centre is 3 units. Hence, if a circle is drawn with centre $z = 1$ and radius 3, then within the circle $|z-1| = 3$, the given function $f(z)$ is analytic and therefore, it can be expanded in a Taylor's series within the circle $|z-1|=3$, which is therefore the circle of convergence.

$$\begin{aligned} \text{By partial fractions, } f(z) &= \frac{z}{z+2} = 1 - \frac{2}{z+2} = 1 - \frac{2}{(z+1)+3} = 1 - \frac{2}{3} \left(1 + \frac{z-1}{3} \right)^{-1} \\ &= 1 - \frac{2}{3} \left[1 - \frac{z-1}{3} + \left(\frac{z-1}{3} \right)^2 - \left(\frac{z-1}{3} \right)^3 + \dots \right]. \end{aligned}$$

Example 28 : Expand $f(z) = \frac{2}{(2z+1)^3}$ about (i) $z = 0$ (ii) $z = 2$.

[JNTU Nov. 2008S (Set No. 4)]

Solution : (i) About $z = 0$

From binomial series, we have

$$\frac{1}{(1-z)^m} = \sum_{n=0}^{\infty} \frac{(m+n-1)!}{(m-1)! n!} z^n, \quad |z| < 1 \quad \text{where } m \text{ is a positive integer.}$$

For $m = 3$, we have

$$\begin{aligned} \frac{1}{(1+2z)^3} &= \sum_{n=0}^{\infty} (-1)^n \frac{(3+n-1)!}{2! n!} (2z)^n, \quad \text{if } |2z| < 1 \\ &= \sum_{n=0}^{\infty} (-1)^n \frac{(n+2)!}{2 \cdot n!} \cdot 2^n z^n \\ &= \sum_{n=0}^{\infty} (-1)^n \frac{(n+2)(n+1)n!}{2 \cdot n!} \cdot 2^n z^n \\ &= \sum_{n=0}^{\infty} (-1)^n (n+1)(n+2) \cdot 2^n z^n, \quad \text{if } |z| < \frac{1}{2} \end{aligned}$$

(ii) About $z = 2$

$$\frac{1}{(1+2z)^3} = \frac{1}{(1+2z-4+4)^3} = \frac{1}{[5+2(z-2)]^3} = \frac{1}{5^3 \left[1 + \frac{2}{5}(z-2) \right]^3}$$

$$= \frac{1}{125} \sum_{n=0}^{\infty} (-1)^n \frac{1}{2} (n+1)(n+2) \cdot \left(\frac{2}{5}\right)^n (z-2)^n \text{ using (1),}$$

$$\text{if } \left| \frac{2}{5} (z-2) \right| < 1 \quad \text{i.e., } |z-2| < \frac{5}{2}$$

5.4. LAURENT'S SERIES EXPANSION

We have seen under Taylor's series that if $f(z)$ is analytic at $z = a$, we can have a series expansion of $f(z)$ in non-negative powers of $z - a$ which is valid in a region given by $|z - a| < R$ for a suitable R . A natural question that arises is: "can we think of having a power series expansion around $z = a$ even if it is a singular point of $f(z)$?" Some of the examples worked out earlier do indicate that such expansions are possible. However, in this case, the region of validity of the expansion is to be carefully written. [See examples (2), (7), (8)]

Laurent's theorem gives a procedure to expand a given function in powers of $(z - a)$. The series expansion may have positive as well as negative powers.

5.5. LAURENT'S THEOREM

[JNTU April 2006, 2007S, Aug. 2007S, 2008S, Nov. 2008S (Set No. 3),
JNTU (H) Nov. 2009 (Set No. 2)]

Let c_1 and c_2 be two circles given by $|z' - a| = r_1$ and $|z' - a| = r_2$ respectively where $r_2 < r_1$ and z' is any point on c_1 or c_2 .

Let $f(z)$ be analytic on c_1 and c_2 and throughout the region between the two circles. Let z be any point in the ring shaped region between the two circles c_1 and c_2 .

$$\text{Then} \quad f(z) = \sum_{n=0}^{\infty} a_n (z-a)^n + \sum_{n=1}^{\infty} \frac{b_n}{(z-a)^n} \quad \dots(1)$$

$$\text{where} \quad a_n = \frac{1}{2\pi i} \oint_{c_1} \frac{f(z')}{(z' - a)^{n+1}} dz' \quad \dots(2)$$

$$b_n = \frac{1}{2\pi i} \oint_{c_2} \frac{f(z')}{(z' - a)^{-n+1}} dz' \quad \dots(3)$$

where the integrals are taken around c_1 and c_2 in the anti-clockwise sense. [Then series in (1) is called the Laurent's series expansion of $f(z)$ around $z = z_0$.]

Proof. Introduce a cross cut AB as in figure below :

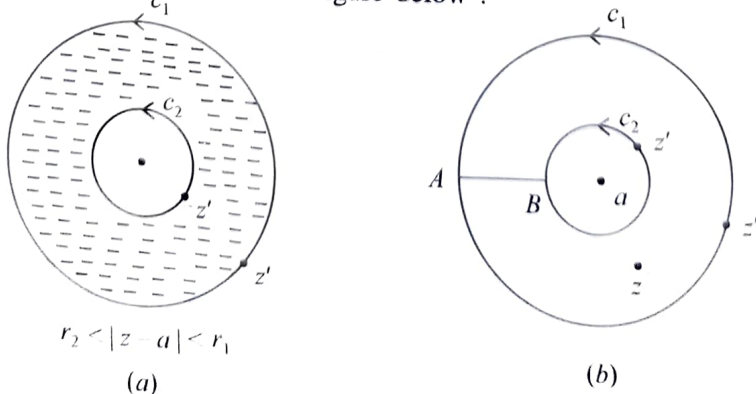


Fig. 5.2

We redraw Fig. (b) as in Fig (c) :

Let B be the positively oriented boundary of the shaded region. We notice that $f(z')$ is analytic in the domain bounded by AB and c_2 , taken in clockwise sense, BA and c_1 taken in anti-clockwise sense. Hence, z is a point in this region of analyticity, by Cauchy's integral formula.

We have

$$\begin{aligned} f(z) &= \frac{1}{2\pi i} \int_B \frac{f(z')}{(z'-z)} dz' \\ &= \frac{1}{2\pi i} \int_{AB} \frac{f(z')}{z'-z} dz' + \frac{1}{2\pi i} \int_{(-c_2)} \frac{f(z')}{z'-z} dz' + \frac{1}{2\pi i} \int_{BA} \frac{f(z')}{(z'-z)} dz' + \frac{1}{2\pi i} \int_{c_1} \frac{f(z')}{z'-z} dz' \\ &= \frac{1}{2\pi i} \oint_{c_1} \frac{f(z')}{z'-z} dz' - \frac{1}{2\pi i} \oint_{c_2} \frac{f(z')}{z'-z} dz' \quad \dots(4) \end{aligned}$$

taking both the integrals in the anti-clockwise sense. ($\int_{AB}$ and $\int_{BA}$ cancel with one another).

Consider $\frac{1}{2\pi i} \int_{c_1} \frac{f(z')}{(z'-z)} dz'$ in (4). As in the proof of Taylor's theorem, we observe that

$$\begin{aligned} \frac{1}{z'-z} &= \frac{1}{(z'-a)-(z-a)} \\ &= \frac{1}{z'-a} + \frac{z-a}{(z'-a)^2} + \frac{(z-a)^2}{(z'-a)^3} + \dots + \frac{(z-a)^{n-1}}{(z'-a)^n} + \frac{(z'-a)^n}{(z-a)^n(z'-z)} \end{aligned}$$

Multiplying both sides by $f(z') / 2\pi i$ and integrating around c_1 with z' as a variable point on c_1 , we get

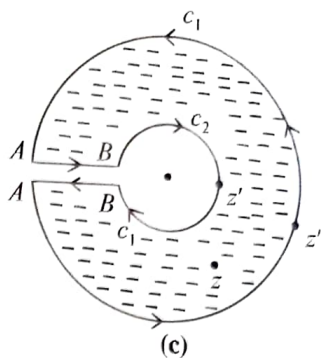
$$\frac{1}{2\pi i} \int_{c_1} \frac{f(z')}{z'-z} dz' = a_0 + a_1(z-a) + a_2(z-a)^2 + \dots + a_{n-1}(z-a)^{n-1} + R_n \quad \dots(5)$$

$$\text{where } a_n = \frac{1}{2\pi i} \int_{c_1} \frac{f(z')}{(z'-a)^{n+1}} dz' \quad [\text{as in (2)}] \quad \dots(6)$$

$$\text{and } R_n = \frac{(z-a)^n}{2\pi i} \int_{c_1} \frac{f(z')}{(z'-a)^n(z'-z)} dz' \quad \dots(7)$$

Consider $\left(\frac{-1}{2\pi i}\right) \int_{c_2} \frac{f(z')}{z'-z} dz'$ in (4). We notice that

$$\begin{aligned} -\frac{1}{z'-z} &= \frac{1}{z-z'} = \frac{1}{(z-a)-(z'-a)} = \frac{1}{(z-a) \left[1 - \frac{z'-a}{z-a}\right]} \\ &= \frac{1}{z-a} + \frac{z'-a}{(z-a)^2} + \frac{(z'-a)^2}{(z-a)^3} + \dots + \frac{(z'-a)^{n-1}}{(z-a)^n} + \frac{(z'-a)^n}{(z-a)^n(z-z')} \end{aligned}$$



Multiplying both sides by $f(z') / 2\pi i$ and integrating around c_2 in anticlockwise sense, we get

$$-\frac{1}{2\pi i} \int_{c_2} \frac{f(z')}{z' - z} dz' = \frac{b_1}{z - a} + \frac{b_2}{(z - a)^2} + \frac{b_3}{(z - a)^3} + \dots + \frac{b_n}{(z - a)^n} + Q_n \quad \dots(8)$$

where $b_n = \frac{1}{2\pi i} \int_{c_2} f(z') (z' - a)^{n-1} dz' = \frac{1}{2\pi i} \int_{c_2} \frac{f(z')}{(z' - a)^{-n+1}} dz' \quad \dots(9)$

and $Q_n = \frac{1}{2\pi i} \int_{c_2} \frac{f(z') (z' - a)^n}{(z - a)^n (z - z')} dz' \quad \dots(10)$

It can be proved that as $n \rightarrow \infty$, R_n and $Q_n \rightarrow 0$.

In view of this, from (4), (5), (6), (8), (9), (10), we get

$$f(z) = \sum_{n=0}^{\infty} a_n (z - a)^n + \sum_{n=1}^{\infty} \frac{b_n}{(z - a)^n}$$

where a_n, b_n are as given in (2), (3).

Hence, Laurent's theorem follows.

The part $\sum_{n=0}^{\infty} a_n (z - a)^n$ is called the **analytic part** while the part $\sum_{n=1}^{\infty} b_n (z - a)^{-n}$ containing negative powers of $(z - a)$ is called the **principal part** of $f(z)$ at the singular point $z = a$.

Note. The proofs of Taylor's theorem and Laurent's theorem are elementary, but lengthy. We have used

$$\frac{1}{1 - \alpha} = 1 + \alpha + \alpha^2 + \dots + \alpha^{n-1} + \frac{\alpha^n}{1 - \alpha}$$

At another stage we multiplied both sides of an equation by $f(z') / 2\pi i$ and integrated over either c_1 or c_2 . The expressions to be written are, of course, lengthy.

The students are advised to write and rewrite the proofs in case they want to master them.

Note. Let us look at the region $r_1 < |z - a| < r_2$.

Since $f(z)$ is analytic in this ring shaped region, if c is any circle like $|z - a| = r$ as in fig. (d) with centre a or any other closed contour enclosing a as in fig. (e).

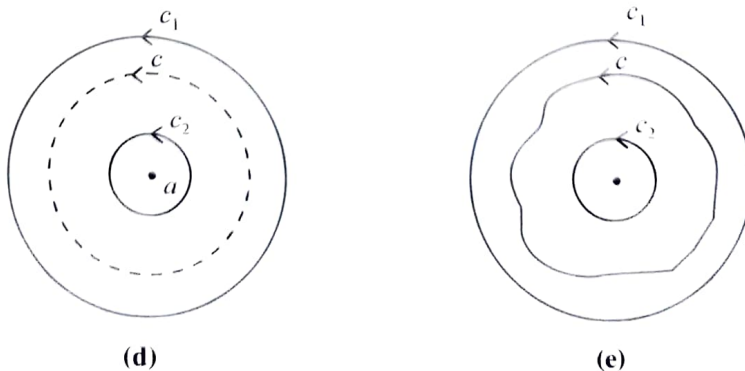


Fig. 5.2

$$\int_{c_1} \frac{f(z')}{(z'-a)^{n+1}} dz' = \int_c \frac{f(z')}{(z'-a)^{n+1}} dz'$$

and

$$\int_{c_2} \frac{f(z')}{(z'-a)^{-n+1}} dz' = \int_c \frac{f(z')}{(z'-a)^{-n+1}} dz'$$

[Using the Note following Cauchy-Goursat theorem for multiply connected regions.]

Hence, $a_n = \frac{1}{2\pi i} \oint_c \frac{f(z')}{(z'-a)^{n+1}} dz'$; $b_n = \frac{1}{2\pi i} \oint_c \frac{f(z')}{(z'-a)^{-n+1}} dz'$

Since the curve c is same in both the above integrals, we notice that b_n is same as a_n where n is replaced by $(-n)$.

Thus, we may restate Laurent's theorem as below :

Let $f(z)$ be analytic in the ring shaped region $r_2 < |z - a| < r_1$.

Let c be any simple closed curve around $z = a$ in the region of analyticity as in figure.

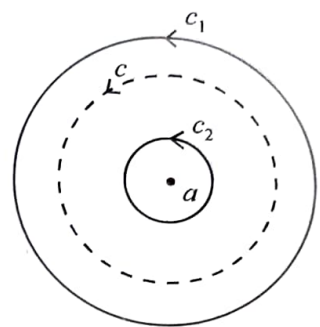


Fig. 5.3

Then, $f(z) = \sum_{n=-\infty}^{\infty} a_n (z-a)^n$... (1)

where $a_n = \frac{1}{2\pi i} \oint_c \frac{f(z')}{(z'-a)^{n+1}} dz'$... (2)

the integration being carried out around c in the anti-clockwise sense.

Note : 1. Here in (2), we can have $a_n = \frac{1}{2\pi i} \oint_c \frac{f(z)}{(z-a)^{n+1}} dz$

Further it will be convenient to take c as a circle $|z - a| = r$ where $r_2 < r < r_1$.

2. The students are advised to understand and assimilate the essence of Taylor's theorem and Laurent's theorem.

If $f(z)$ is analytic on and within c_1 everywhere (including $z = a$), the Laurent's series expansion of $f(z)$ in powers of $z - a$ is just its Taylor's series expansion.

If $f(z)$ is analytic in $r_2 < |z - a| < r_1$ and there are some singular points of $f(z)$ within c_2 , Laurent series and Taylor's series around $z = a$ will be in general different [even if $z = a$ is a point of analyticity for $f(z)$.]

We shall present some examples to explain the procedure to obtain Laurent's series expansion of $f(z)$.

SOLVED EXAMPLES

Example 1 : Let $f(z) = \frac{1}{(1-z)(z-2)}$

Find (a) Maclaurin's series expansion of $f(z)$.

(b) Laurent's series expansion in the annulus region $1 < |z| < 2$.

(c) Laurent's series expansion in $|z| > 2$.

Solution : $f(z) = \frac{1}{(1-z)(z-2)} = -\frac{1}{(z-1)(z-2)}$

$$= \frac{(z-2) - (z-1)}{(z-1)(z-2)} = \frac{1}{(z-1)} - \frac{1}{z-2}$$

The singular points of $f(z)$ are $z = 1$, $z = 2$, $z = 0$ is a point of analyticity. The nearest singular point to $z = 0$ is $z = 1$. Hence, the Maclaurin's series expansion of $f(z)$ is valid in $|z| < 1$.

(a) Maclaurin's series expansion of $f(z)$

$$f(z) = \frac{1}{z-1} - \frac{1}{z-2} = \frac{-1}{1-z} + \frac{1}{2-z} = \frac{-1}{1-z} + \frac{1}{2\left(1-\frac{z}{2}\right)}$$

$$= -\left[1 + z + z^2 + z^3 + \dots\right] + \frac{1}{2}\left[1 + \frac{z}{2} + \left(\frac{z}{2}\right)^2 + \left(\frac{z}{2}\right)^3 + \dots\right]$$

[Here the first expansion is valid for $|z| < 1$ and the second one is valid for

$\left|\frac{z}{2}\right| < 1$, i.e., $|z| < 2$; hence, both are valid in the common region $|z| < 1$.]

Hence,

$$f(z) = -\sum_{n=0}^{\infty} z^n + \frac{1}{2} \sum_{n=0}^{\infty} \left(\frac{z}{2}\right)^n$$

$$= \sum_{n=0}^{\infty} \left[2^{-(n+1)} - 1\right] z^n \quad \text{if } |z| < 1.$$

This is the Maclaurin's series expansion of $f(z)$.

(b) Laurent's series expansion in $1 < |z| < 2$.

$|z| = 1$, $|z| = 2$ are two concentric circles with centre at o and radii equal to 1 and 2 respectively. In $1 < |z| < 2$, $f(z)$ is analytic

$$f(z) = \frac{1}{z-1} - \frac{1}{z-2} = \frac{1}{z\left(1-\frac{1}{z}\right)} + \frac{1}{2\left(1-\frac{z}{2}\right)}$$

$$= \frac{1}{z}\left[1 + \frac{1}{z} + \frac{1}{z^2} + \frac{1}{z^3} + \dots\right] + \frac{1}{2}\left[1 + \frac{z}{2} + \left(\frac{z}{2}\right)^2 + \left(\frac{z}{2}\right)^3 + \dots\right]$$

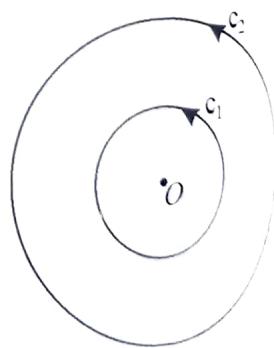


Fig. 5.4

Here the first expansion is valid if $\left| \frac{1}{z} \right| < 1$, i.e., if $|z| > 1$ and the second expansion is valid if $\left| \frac{z}{2} \right| < 1$, i.e., if $|z| < 2$.

$$\text{Hence, } f(z) = \left(\frac{1}{z} + \frac{1}{z^2} + \frac{1}{z^3} + \frac{1}{z^4} + \dots \right) + \left(\frac{1}{2} + \frac{z}{2^2} + \frac{z}{2^3} + \frac{z}{2^4} + \dots \right) = \sum_{n=-\infty}^{\infty} a_n z^n$$

$$\text{where } a_n = \begin{cases} \frac{1}{2^{n+1}}, & \text{if } n = 0, 1, 2, \dots \\ 1, & \text{if } n = -1, -2, -3, \dots \end{cases}$$

The expansion is valid if both $|z| > 1$ and $|z| < 2$ are true, i.e., if $1 < |z| < 2$.

(c) **Laurent's series expansion for $|z| > 2$.**

$$\begin{aligned} f(z) &= \frac{1}{z-1} - \frac{1}{z-2} = \frac{1}{z\left(1-\frac{1}{z}\right)} - \frac{1}{z\left(1-\frac{2}{z}\right)} \\ &= \frac{1}{z} \left[1 + \frac{1}{z} + \frac{1}{z^2} + \frac{1}{z^3} + \dots \right] - \frac{1}{z} \left[1 + \frac{2}{z} + \left(\frac{2}{z}\right)^2 + \left(\frac{2}{z}\right)^3 + \dots \right] \text{ if } \left| \frac{1}{z} \right| < 1 \text{ and } \left| \frac{2}{z} \right| < 1 \\ &= \sum_{n=1}^{\infty} \frac{(1-2^n)}{z^{n+1}} \text{ i.e., if } |z| > 1 \text{ and } |z| > 2 \end{aligned}$$

$$\text{Hence, } f(z) = \sum_{n=1}^{\infty} \frac{(1-2^n)}{z^{n+1}} \text{ if } |z| > 2.$$

Example 2 : Represent $f(z) = \frac{z+1}{z-1}$ by

- Maclaurin's series giving the region of validity and
- Laurent's series for the domain $|z| > 1$.

Solution : We have $f(z) = \frac{z+1}{z-1} = \frac{z-1+2}{z-1} = 1 + \frac{2}{z-1}$

- $z = 1$ is the only singular point of $f(z)$. $f(z)$ is analytic at $z = 0$.

[Hence, Maclaurin's series expansion of $f(z)$ is valid on $|z| < 1$.]

We have

$$\begin{aligned} f(z) &= 1 - \frac{2}{1-z} = 1 - 2(1 + z + z^2 + z^3 \dots) \text{ if } |z| < 1 \\ &= -1 - 2 \sum_{n=1}^{\infty} z^n \text{ if } |z| < 1. \end{aligned}$$

This is the required Maclaurin's series expansion.

Handwritten notes on the right margin:

- $1 > \frac{2}{|z|} > \frac{1}{|z|}$
- $z = 2$
- $z(1 - \frac{2}{z})$
- $1 < |z|$
- $\frac{1}{|z|} < 1$

$$\begin{aligned}
 (b) \quad f(z) &= 1 + \frac{2}{z-1} = 1 + \frac{2}{z \left(1 - \frac{1}{z}\right)} = 1 + \frac{2}{z} \left(1 - \frac{1}{z}\right)^{-1} \\
 &= 1 + \frac{2}{z} \left[1 + \frac{1}{z} + \frac{1}{z^2} + \dots\right] \text{ if } \left|\frac{1}{z}\right| < 1 \\
 &= 1 + 2 \sum_{n=1}^{\infty} \frac{1}{z^n} \text{ if } |z| > 1.
 \end{aligned}$$

This is the required Laurent's series expansion.

Example 3 : Give two Laurent's series expansions in powers of z for $f(z) = \frac{1}{z^2(1-z)}$ and specify the regions in which these expansions are valid.

[JNTU Nov. 2008, JNTU (A) Dec. 2009 (Set No. 2)]

(or) Find Laurent's series for $f(z) = \frac{1}{z^2(1-z)}$ and find the region of convergence.

[JNTU April 2006 (Set No. 1)]

Solution : Given $f(z) = \frac{1}{z^2(1-z)}$, its singular points are $z = 0$ and $z = 1$.

$$\begin{aligned}
 \text{Now } f(z) &= \frac{1}{z^2(1-z)} = \frac{1}{z^2} \cdot \frac{1}{(1-z)} = \frac{1}{z^2} (1-z)^{-1} \\
 &= \frac{1}{z^2} [1 + z + z^2 + \dots] \text{ if } z \neq 0 \text{ and } |z| < 1 \\
 &= \frac{1}{z^2} + \frac{1}{z} + 1 + z + z^2 + \dots \\
 &= \sum_{n=0}^{\infty} z^{n-2} \text{ if } 0 < |z| < 1.
 \end{aligned}$$

This is one Laurent's series expansion in powers of z .
Also

$$\begin{aligned}
 f(z) &= \frac{1}{z^2(1-z)} = \frac{-1}{z^2(z-1)} \\
 &= \frac{-1}{z^2 \cdot z \left(1 - \frac{1}{z}\right)} = \frac{-1}{z^3 \left(1 - \frac{1}{z}\right)} = \frac{-1}{z^3} \left(1 - \frac{1}{z}\right)^{-1} \\
 &= \frac{-1}{z^3} \left(1 + \frac{1}{z} + \frac{1}{z^2} + \dots\right) \text{ if } z \neq 0 \text{ and } \left|\frac{1}{z}\right| < 1 \\
 &= -\left(\frac{1}{z^3} + \frac{1}{z^4} + \frac{1}{z^5} + \dots\right) \text{ if } |z| > 1 \\
 &= -\sum_{n=0}^{\infty} z^{-n-3} \text{ if } |z| > 1
 \end{aligned}$$

This is another Laurent's series expansion in powers of z .

Example 4 : Expand $f(z) = \frac{1}{z^2 - 3z + 2}$ in the region (i) $0 < |z - 1| < 1$ (ii) $1 < |z| < 2$
 [JNTU 2003S, 2004S, April 2006, Aug. 2007S, 2007S, Nov. 2008S, Nov. 2008,

JNTU (A), (H) Nov. 2009 (Set No. 1,2)]

Solution : $z = 1, z = 2$ are the singular points of $f(z)$.

We have

$$\begin{aligned} f(z) &= \frac{1}{z^2 - 3z + 2} = \frac{1}{(z-1)(z-2)} \\ &= \frac{(z-1) - (z-2)}{(z-1)(z-2)} = \frac{1}{z-2} - \frac{1}{z-1} \end{aligned}$$

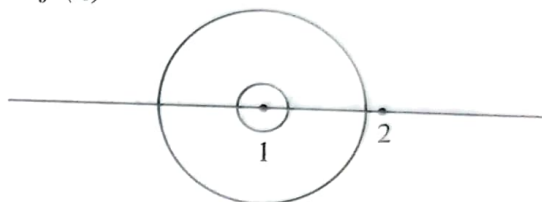


Fig. 5.5

(i) The function $f(z)$ is analytic in the ring shaped region $0 < r_2 < |z - 1| < r_1 < 1$.
 Put $z - 1 = w \quad \therefore \quad z = w + 1 \quad \therefore \quad z - 2 = w + 1 - 2 = w - 1$

$$\begin{aligned} \therefore f(z) &= \frac{1}{w-1} - \frac{1}{w} = -\frac{1}{w} - \frac{1}{1-w} \\ &= -\frac{1}{w} - [1 + w + w^2 + \dots] \text{ if } w < 1 \text{ and } w \neq 0 \\ &= -\frac{1}{z-1} - \sum_{n=0}^{\infty} (z-1)^n \text{ if } 0 < |z-1| < 1 \\ &= -\sum_{n=-1}^{\infty} (z-1)^n \text{ if } 0 < |z-1| < 1. \end{aligned}$$

$0 < |z-1| < 1$



(ii) Given $1 < |z| < 2$ i.e., $1 < |z|$ and $|z| < 2$. or $\left|\frac{1}{z}\right| < 1$ and $\left|\frac{z}{2}\right| < 1$

$$\begin{aligned} \therefore f(z) &= \frac{1}{z-2} - \frac{1}{z-1} = \frac{1}{-2\left(1-\frac{z}{2}\right)} - \frac{1}{z\left(1-\frac{1}{z}\right)} = -\frac{1}{2}\left(1-\frac{z}{2}\right)^{-1} - \frac{1}{z}\left(1-\frac{1}{z}\right)^{-1} \\ &= -\frac{1}{2}\left[1 + \frac{z}{2} + \frac{z^2}{4} + \frac{z^3}{8} + \dots\right] - \frac{1}{z}\left[1 + \frac{1}{z} + \frac{1}{z^2} + \frac{1}{z^3} + \dots\right] \\ &= -\frac{1}{2}\left[1 + \frac{z}{2} + \frac{z^2}{4} + \frac{z^3}{8} + \dots\right] - \left[\frac{1}{z} + \frac{1}{z^2} + \frac{1}{z^3} + \dots\right] \end{aligned}$$

Example 5 : Expand the function $f(z) = \frac{z-1}{z}$ as Laurent series for $|z - 1| > 1$.

Solution : Write $z - 1 = w \quad \therefore \quad z = 1 + w$.

$$\therefore f(z) = \frac{w}{1+w} = \frac{w}{w\left(1+\frac{1}{w}\right)} = \frac{1}{1+\frac{1}{w}}$$

$$= 1 - \frac{1}{w} + \frac{1}{w^2} - \frac{1}{w^3} + \dots \text{ if } \left| \frac{1}{w} \right| < 1$$

$$= \sum_{n=0}^{\infty} \frac{(-1)^n}{w^n} \text{ if } |w| > 1 = \sum_{n=0}^{\infty} \frac{(-1)^n}{(z-1)^n} \text{ if } |z-1| > 1.$$

This is the required expansion.

Example 6 : Expand $f(z) = \frac{e^{2z}}{(z-1)^3}$ about $z = 1$ as a Laurent's series. Also find the region

of convergence.

[JNTU 2002, Aug. 2007S, 2008S (Set No. 3)]

Solution : $f(z) = \frac{e^{2z}}{(z-1)^3}.$

We want the Laurent's series expansion of $f(z)$ around $z = 1$.

Put $z - 1 = w. \quad \therefore z = w + 1.$

$$\begin{aligned} \therefore f(z) &= \frac{e^{2(1+w)}}{w^3} = e^2 \cdot \frac{e^{2w}}{w^3} \\ &= e^2 \cdot \frac{1}{w^3} \left[1 + (2w) + \frac{(2w)^2}{2!} + \frac{(2w)^3}{3!} + \dots \right] \\ &= e^2 \sum_{n=0}^{\infty} \frac{2^n}{n!} (w^{n-3}) \text{ if } w \neq 0 \\ &= e^2 \sum_{n=0}^{\infty} \frac{2^n}{n!} (z-1)^{n-3} \text{ if } z-1 \neq 0 \\ &= e^2 \sum_{n=0}^{\infty} \frac{2^n}{n!} (z-1)^{n-3} \text{ if } |z-1| > 0 \end{aligned}$$

Example 7 : Find the Laurent series expansion of the function $f(z) = \frac{z^2 - 6z - 1}{(z-1)(z-3)(z+2)}$ in the region $3 < |z+2| < 5$.

[JNTU 2001, 2005S, Nov. 2006, Aug. 2007S, Nov. 2008S (Set No. 2)]

Solution : By partial fractions

$$\begin{aligned} \frac{z^2 - 6z - 1}{(z-1)(z-3)(z+2)} &= \frac{1}{z-1} - \frac{1}{z-3} + \frac{1}{z+2} = \frac{1}{z+2-3} - \frac{1}{z+2-5} + \frac{1}{z+2} \\ &= \frac{1}{(z+2)\left(1 - \frac{3}{z+2}\right)} + \frac{1}{5\left(1 - \frac{z+2}{5}\right)} + \frac{1}{z+2} \\ &= \frac{1}{z+2} \sum_{n=0}^{\infty} \left(\frac{3}{z+2}\right)^n + \frac{1}{5} \sum_{n=0}^{\infty} \left(\frac{z+2}{5}\right)^n + \frac{1}{z+2} \end{aligned}$$

valid for $\left| \frac{3}{z+2} \right| < 1$ and $\left| \frac{z+2}{5} \right| < 1 \Rightarrow 3 < |z+2| < 5$

Example 8 : Expand $f(z) = \frac{z}{z^2+1}$ for $|z-3| > 2$.

[JNTU 2000]

Solution : Using partial fractions

$$\begin{aligned} \frac{z}{z^2+1} &= \frac{z}{(z+i)(z-i)} = \frac{A}{z+i} + \frac{B}{z-i} = \frac{1}{2} \left[\frac{1}{z+i} + \frac{1}{z-i} \right] \\ &= \frac{1}{2} \left[\frac{1}{z-3+3-i} + \frac{1}{z-3+3-i} \right] \\ &= \frac{1}{2(z-3)} \left[\frac{1}{1+\frac{3+i}{z-3}} + \frac{1}{1+\frac{3-i}{z-3}} \right] = \frac{1}{2(z-3)} \sum_{n=0}^{\infty} (-1)^n \left\{ \left(\frac{3+i}{z-3} \right)^n + \left(\frac{3-i}{z-3} \right)^n \right\} \end{aligned}$$

valid for $\left| \frac{3+i}{z-3} \right| < 1$ and $\left| \frac{3-i}{z-3} \right| < 1$ i.e. $|z-3| > |3 \pm i| = \sqrt{9+1} = \sqrt{10} = 3.16$.

Example 9 : Find the Laurent series of $\frac{7z-2}{(z+1)z(z-2)}$ in the annulus $1 < |z+1| < 3$.

[JNTU 2003S (Set No. 4), 2004S]

Solution : Let $f(z) = \frac{7z-2}{(z+1)z(z-2)}$

Put $z+1=w$. Then $z=w-1$

$$\therefore f(z) = \frac{7(w-1)-2}{w(w-1)(w-1-2)} = \frac{7w-9}{w(w-1)(w-3)}$$

By partial fractions

$$\begin{aligned} f(z) &= \frac{A}{w} + \frac{B}{w-1} + \frac{C}{w-3} = -\frac{3}{w} + \frac{1}{w-1} + \frac{2}{w-3} \\ &= -\frac{3}{w} + \frac{1}{w \left(1 - \frac{1}{w} \right)} - \frac{2}{3 \left(1 - \frac{w}{3} \right)} = -\frac{3}{w} + \frac{1}{w} \left(1 - \frac{1}{w} \right)^{-1} - \frac{2}{3} \left(1 - \frac{w}{3} \right)^{-1} \\ &= -\frac{3}{w} + \frac{1}{w} \left(1 + \frac{1}{w} + \frac{1}{w^2} + \dots \right) - \frac{2}{3} \left(1 + \frac{w}{3} + \frac{w^2}{3^2} + \dots \right) \\ &= \left(-\frac{2}{w} + \frac{1}{w^2} + \frac{1}{w^3} + \dots \right) - \frac{2}{3} \left(1 + \frac{w}{3} + \frac{w^2}{3^2} + \dots \right) \end{aligned}$$

$$= -\frac{2}{z+1} + \frac{1}{(z+1)^2} + \frac{1}{(z+1)^3} + \dots - \frac{2}{3} \left[1 + \frac{z+1}{3} + \frac{(z+1)^2}{3^2} + \dots \right]$$

The above series is valid in the region

$$\left| \frac{1}{w} \right| < 1 \quad \text{and} \quad \left| \frac{w}{3} \right| < 1$$

i.e., $1 < |w|$ and $|w| < 3$ i.e., $1 < |w| < 3$ or $1 < |z+1| < 3$

Example 10 : Expand the Laurent series of $\frac{z^2-1}{(z+2)(z+3)}$, for $|z| > 3$. $3 < |z|$
 $\frac{3}{2} < 1$

[JNTU 2004, April 2006, 2008S (Set No. 1)]

Solution : Let $f(z) = \frac{z^2-1}{(z+2)(z+3)} = 1 - \frac{5z+7}{(z+2)(z+3)}$ [By actual division]

$$= 1 + \frac{3}{z+2} - \frac{8}{z+3} \quad (\text{Resolving into partial fractions})$$

Given $|z| > 3 \Rightarrow \frac{3}{|z|} < 1$ or $\left| \frac{3}{z} \right| < 1$. Also $\left| \frac{2}{z} \right| < 1$ $\left(\frac{3}{2} \right) < 1$
 $\left(\frac{2}{z} \right) < 1$

$$\therefore f(z) = 1 + \frac{3}{z+2} - \frac{8}{z+3}$$

$$\begin{aligned} &= 1 + \frac{3}{z \left(1 + \frac{2}{z} \right)} - \frac{8}{z \left(1 + \frac{3}{z} \right)} = 1 + \frac{3}{z} \left(1 + \frac{2}{z} \right)^{-1} - \frac{8}{z} \left(1 + \frac{3}{z} \right)^{-1} \\ &= 1 + \frac{3}{z} \left[1 - \frac{2}{z} + \left(\frac{2}{z} \right)^2 - \left(\frac{2}{z} \right)^3 + \dots \right] - \frac{8}{z} \left[1 - \frac{3}{z} + \left(\frac{3}{z} \right)^2 - \left(\frac{3}{z} \right)^3 + \dots \right] \\ &= 1 + \frac{3}{z} \sum_{n=0}^{\infty} (-1)^n \left(\frac{2}{z} \right)^n - \frac{8}{z} \sum_{n=0}^{\infty} (-1)^n \left(\frac{3}{z} \right)^n \end{aligned}$$

which is the required Laurent series.

Example 11 : Expand $f(z) = \frac{z+3}{z(z^2-z-2)}$ in powers of z where

(i) $|z| < 1$ (ii) $1 < |z| < 2$ (iii) $|z| > 2$

Solution : By partial fractions

$$\begin{aligned} f(z) &= \frac{z+3}{z(z^2-z-2)} = \frac{z+3}{z(z+1)(z-2)} \\ &= -\frac{3}{2} \cdot \frac{1}{z} + \frac{2}{3} \cdot \frac{1}{z+1} + \frac{5}{6} \cdot \frac{1}{z-2} \end{aligned}$$

[JNTU 2004]

(i) For $|z| < 1$, we write (1) as $f(z) = -\frac{3}{2} \cdot \frac{1}{z} + \frac{2}{3} \cdot \frac{1}{1+z} + \frac{5}{6} \cdot \frac{1}{-2\left(1-\frac{z}{2}\right)}$

Notice that both $\left|\frac{z}{2}\right|$ and $|z|$ are less than 1.

$$\begin{aligned}\therefore f(z) &= -\frac{3}{2} \cdot \frac{1}{z} + \frac{2}{3}(1+z)^{-1} - \frac{5}{12}\left(1-\frac{z}{2}\right)^{-1} \\ &= -\frac{3}{2z} + \frac{2}{3}(1-z+z^2-z^3+\dots) - \frac{5}{12}\left(1+\frac{z}{2}+\frac{z^2}{2^2}+\frac{z^3}{2^3}+\dots\right) \\ &= -\frac{3}{2z} + \frac{1}{4} - \frac{7}{8}z + \frac{9}{16}z^2 - \frac{23}{32}z^3 + \dots\end{aligned}$$

(ii) For $1 < |z| < 2$, we write (1) as

$$f(z) = -\frac{3}{2} \cdot \frac{1}{z} + \frac{2}{3} \cdot \frac{1}{z\left(1+\frac{1}{z}\right)} + \frac{5}{6} \cdot \frac{1}{2\left(\frac{z}{2}-1\right)}$$

and notice that both $|1/z|$ and $|z/2|$ are less than 1.

$$\begin{aligned}\therefore f(z) &= -\frac{3}{2z} + \frac{2}{3z}\left(1+\frac{1}{z}\right)^{-1} - \frac{5}{12}\left(1-\frac{z}{2}\right)^{-1} \\ &= -\frac{3}{2z} + \frac{2}{3z}\left(1-\frac{1}{z}+\frac{1}{z^2}-\frac{1}{z^3}+\dots\right) - \frac{5}{12}\left(1+\frac{z}{2}+\frac{z^2}{2^2}+\frac{z^3}{2^3}+\dots\right) \\ &= -\frac{5}{12} - \frac{5}{6z} + \frac{2}{3}\left(-\frac{1}{z^2}+\frac{1}{z^3}-\frac{1}{z^4}+\dots\right) - \frac{5}{12}\left(\frac{z}{2}+\frac{z^2}{4}+\frac{z^3}{8}+\dots\right)\end{aligned}$$

(iii) For $|z| > 2$, we write (1) as

$$\begin{aligned}f(z) &= -\frac{3}{2z} + \frac{2}{3z}\left(1+\frac{1}{z}\right)^{-1} + \frac{5}{6z}\left(1-\frac{2}{z}\right)^{-1} \\ &= -\frac{3}{2z} + \frac{2}{3z}\left(1-\frac{1}{z}+\frac{1}{z^2}-\frac{1}{z^3}+\dots\right) + \frac{5}{6z}\left(1+\frac{2}{z}+\frac{2^2}{z^2}+\frac{2^3}{z^3}+\dots\right) \\ &= \frac{1}{z^2} + \frac{4}{z^3} + \frac{6}{z^4} + \dots\end{aligned}$$

Example 12 : Obtain Laurent's expansion for $f(z) = \frac{1}{(z+2)(1+z)^2}$ in

(i) $|z| < 2$ (ii) $|1+z| > 1$.

[JNTU 2003S (Set No. 2)]

Solution : By partial fractions

$$f(z) = \frac{1}{(z+2)(1+z)^2} = \frac{A}{z+2} + \frac{B}{z+1} + \frac{C}{(z+1)^2} = \frac{1}{z+2} - \frac{1}{z+1} + \frac{1}{(z+1)^2} \quad \dots (1)$$

(i) For $|z| < 2$, we write (1) as $f(z) = \frac{1}{2\left(\frac{z}{2} + 1\right)} - \frac{1}{z+1} + \frac{1}{(z+1)^2}$

Notice that both $\left|\frac{z}{2}\right|$ and $|z|$ are less than 1.

$$\begin{aligned}\therefore f(z) &= \frac{1}{2} \left(1 + \frac{z}{2}\right)^{-1} - (1+z)^{-1} + (1+z)^{-2} \\ &= \frac{1}{2} \left[1 - \frac{z}{2} + \left(\frac{z}{2}\right)^2 - \left(\frac{z}{2}\right)^3 + \dots\right] - (1 - z + z^2 - z^3 + \dots) + (1 - 2z + 3z^2 - 4z^3 + \dots)\end{aligned}$$

(ii) For $|1+z| > 1$, put $1+z = w \quad \therefore \quad z = w-1$

$$\begin{aligned}\therefore f(z) &= \frac{1}{(z+2)(1+z)^2} = \frac{1}{(w+1)w^2} \\ &= \frac{1}{w^2} (1+w)^{-1} = \frac{1}{w^2} \cdot w^{-1} \left(\frac{1}{w} + 1\right)^{-1} = \frac{1}{w} \left(1 + \frac{1}{w}\right)^{-1} \\ &= \frac{1}{w} \left(1 - \frac{1}{w} + \frac{1}{w^2} - \frac{1}{w^3} + \dots\right) = \frac{1}{w} - \frac{1}{w^2} + \frac{1}{w^3} - \frac{1}{w^4} + \dots \\ &= \frac{1}{1+z} - \frac{1}{(1+z)^2} + \frac{1}{(1+z)^3} - \frac{1}{(1+z)^4} + \dots\end{aligned}$$

An important Observation :

If $z = a$ is a point of analyticity for $f(z)$, the Taylor's series expansion is unique and its region of validity can be written down.

We must understand that this is not the case with respect to Laurent's series expansion of $f(z)$ around $z = a$. There can be different Laurent's series expansions of $f(z)$ in powers of $(z-a)$ whose regions of validity are different. We shall illustrate this point through an example.

Take

$$f(z) = \frac{1}{(z-2)(z-3)(z-4)}$$

Its singular points are $z = 2, 3, 4$.

(a) Around $z = 0$, we can have Maclaurin's series expansion of $f(z)$ which is valid for $|z| < 2$

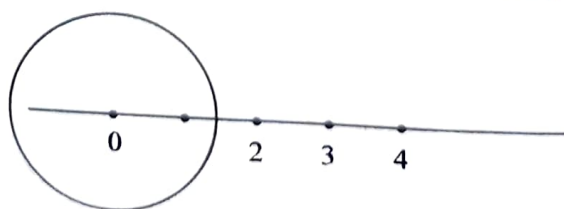


Fig. 5.6

(b) Around $z = 0$, we can have Laurent's series expansion valid for $2 < |z| < 3$.

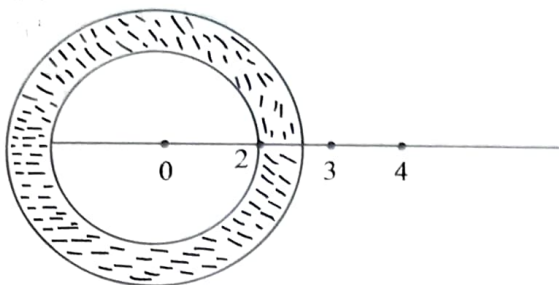


Fig. 5.7

(c) Around $z = 0$, we can have Laurent's series expansion valid for $3 < |z| < 4$.

(d) Around $z = 0$, we can have Laurent's series expansion valid for $|z| > 4$.

(e) Suppose we want expansion of $f(z)$ in powers of $(z - 3)$.

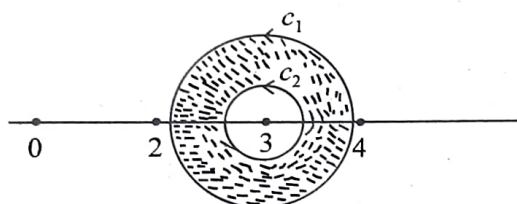


Fig. 5.8

(i) With 3 as centre draw small circle c_2 of radius r_2 . With 3 as centre draw a circle c_1 of radius r_1 . Let these be such that in $r_2 < |z - 3| < r_1$.

$f(z)$ is analytic. For this to happen r_2 can be quite small but +ve and r_1 can be less than 1 only.

Hence, in $0 < |z - 3| < 1$.

We can have Laurent's series expansion in powers of $z - 3$.

(ii) It is also possible to have a circle c_2 with centre at 3 and radius more than 1 and another circle c_1 with centre at 3 and radius less than 3.

i.e., $1 < |z - 3| < 3$.

In this region we can have a valid Laurent's series expansion in powers of $z - 3$.

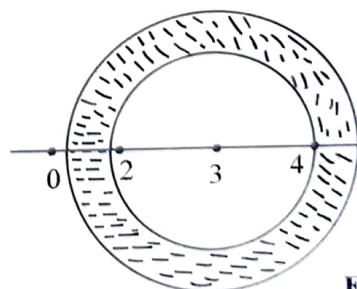


Fig. 5.9

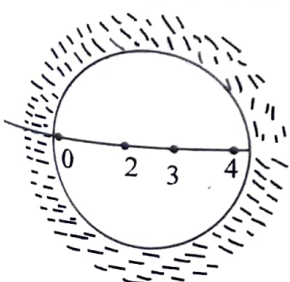
(iii) We can also have, for

$$3 < |z - 3| < R$$

where R can be any large positive number, we can have a valid Laurent's series expansion.

The students are advised to understand the ideas presented carefully.

The Laurent's series plays a crucial part in the development of the theory of residues which will be treated in the next chapter.



$$|z - 3| > 3$$

Fig. 5.10

Example 13 : Obtain all the Laurent series of the function $\frac{7z-2}{(z+1)(z)(z-2)}$ about $z_0 = -1$.
[JNTU 2004, 2007S (Set No. 2)]

Solution : Let $f(z) = \frac{7z-2}{(z+1)(z)(z-2)}$.

This function has 3 singular points at $z_0 = -1, 0, 2$.

By Partial fractions

$$f(z) = \frac{7z-2}{z(z+1)(z-2)} = \frac{A}{z} + \frac{B}{z+1} + \frac{C}{z-2} = \frac{1}{z} - \frac{3}{z+1} + \frac{2}{z-2}$$

Three Laurent series of $f(z)$ can be obtained about $z_0 = -1$ in the regions $0 < |z+1| < 1$, $1 < |z+1| < 3$ and $|z+1| > 3$.

(a) For $0 < |z+1| < 1$

$$\begin{aligned} f(z) &= \frac{1}{(z+1)-1} - \frac{3}{z+1} + \frac{2}{(z+1)-3} = -\frac{1}{1-(z+1)} - \frac{3}{z+1} - \frac{2}{3\left(1-\frac{z+1}{3}\right)} \\ &= (-1)[1-(z+1)]^{-1} - \frac{3}{z+1} - \frac{2}{3}\left[1-\left(\frac{z+1}{3}\right)\right]^{-1} \\ &= (-1)\left[1+(z+1)+(z+1)^2+(z+1)^3+\dots\right] - \frac{3}{z+1} \\ &\quad - \frac{2}{3}\left[1+\frac{z+1}{3}+\frac{(z+1)^2}{3^2}+\frac{(z+1)^3}{3^3}+\dots\right] \\ &= -\frac{3}{z+1} - \frac{5}{3} - \frac{11}{9}(z+1) - \frac{29}{27}(z+1)^2 - \frac{83}{81}(z+1)^3 - \dots \end{aligned}$$

(b) For $1 < |z+1| < 3 \Rightarrow |z+1| > 1$ and $|z+1| < 3 \Rightarrow \frac{1}{|z+1|} < 1$ and $\frac{|z+1|}{3} < 1$

$$\begin{aligned} \therefore f(z) &= \frac{1}{z} - \frac{3}{z+1} + \frac{2}{z-2} = \frac{1}{(z+1)-1} - \frac{3}{z+1} + \frac{2}{(z+1)-3} \\ &= \frac{1}{(z+1)}\left[1-\frac{1}{z+1}\right]^{-1} - \frac{3}{z+1} - \frac{2}{3}\left[1-\frac{z+1}{3}\right]^{-1} \\ &= \frac{1}{(z+1)}\left[1+\frac{1}{z+1}+\frac{1}{(z+1)^2}+\dots\right] - \frac{3}{z+1} - \frac{2}{3}\left[1+\frac{z+1}{3}+\frac{(z+1)^2}{3^2}+\frac{(z+1)^3}{3^3}+\dots\right] \\ &= \left[\frac{1}{z+1}+\frac{1}{(z+1)^2}+\frac{1}{(z+1)^3}+\dots\right] - \frac{3}{z+1} - \frac{2}{3} - \frac{2}{9}(z+1) - \frac{2}{27}(z+1)^2 - \frac{2}{81}(z+1)^3 - \dots \\ &= \left[\frac{1}{(z+1)^2}+\frac{1}{(z+1)^3}+\dots\right] - \frac{2}{z+1} - \frac{2}{3} - \frac{2}{9}(z+1) - \frac{2}{27}(z+1)^2 - \frac{2}{81}(z+1)^3 - \dots \end{aligned}$$

$$(c) \text{ For } |z+1| > 3 \Rightarrow \frac{|z+1|}{3} > 1 \Rightarrow \frac{3}{|z+1|} < 1$$

$$\begin{aligned} \therefore f(z) &= \frac{1}{z} - \frac{3}{z+1} + \frac{2}{z-2} = \frac{1}{(z+1)-1} - \frac{3}{z+1} + \frac{2}{(z+1)-3} \\ &= \frac{1}{(z+1)} \left[1 - \frac{1}{z+1} \right]^{-1} - \frac{3}{z+1} + \frac{2}{z+1} \left[1 - \frac{3}{z+1} \right]^{-1} \\ &= \frac{1}{z+1} \left[1 + \frac{1}{z+1} + \frac{1}{(z+1)^2} + \dots \right] - \frac{3}{z+1} + \frac{2}{z+1} \left[1 + \frac{3}{z+1} + \frac{9}{(z+1)^2} + \dots \right] \\ &= \frac{1}{z+1} + \frac{1}{(z+1)^2} + \frac{1}{(z+1)^3} + \dots - \frac{3}{z+1} + \frac{2}{z+1} + \frac{6}{(z+1)^2} + \frac{18}{(z+1)^3} + \dots \\ &= \frac{7}{(z+1)^2} + \frac{19}{(z+1)^3} + \dots \end{aligned}$$

Example 14 : Obtain the Laurent's series expansion of $f(z) = \frac{e^z}{z(1-z)}$ about $z = 1$.

$$\begin{aligned} \text{Solution: } f(z) &= \frac{e^z}{z(1-z)} = -\frac{e^{z-1} \cdot e}{z(z-1)} = -\frac{e}{z-1} \cdot \frac{e^{z-1}}{(z-1)+1} \\ &= -\frac{e}{z-1} \cdot e^{z-1} \cdot [1 + (z-1)]^{-1} \\ &= -\frac{e}{z-1} \left[\sum_{n=0}^{\infty} \frac{(z-1)^n}{n!} \right] \left[\sum_{n=0}^{\infty} (-1)^n (z-1)^n \right] \text{ if } |z-1| < 1 \\ &= -\frac{e}{z-1} \left[1 + (z-1) + \frac{(z-1)^2}{2!} + \frac{(z-1)^3}{3!} + \dots \right] \left[1 - (z-1) + (z-1)^2 - (z-1)^3 + \dots \right] \\ &= -\frac{e}{z-1} \left[1 - (z-1) + (z-1)^2 + (z-1) - (z-1)^2 + \frac{(z-1)^2}{2} - \dots \right] \\ &= -\frac{e}{z-1} \left[1 - (z-1) + (z-1)^2 - (z-1)^3 + (z-1) - (z-1)^2 + (z-1)^3 - (z-1)^4 \right. \\ &\quad \left. + \frac{(z-1)^2}{2} - \frac{(z-1)^3}{2} + \frac{(z-1)^4}{2} + \frac{(z-1)^3}{6} - \frac{(z-1)^4}{6} + \dots \right] \\ &= -\frac{e}{z-1} \left[1 + \frac{(z-1)^2}{2} - \frac{1}{3}(z-1)^3 - \frac{2}{3}(z-1)^4 - \dots \right] \\ &= e \left[-\frac{1}{z-1} - \frac{1}{2}(z-1) + \frac{1}{3}(z-1)^2 - \frac{2}{3}(z-1)^3 + \dots \right] \end{aligned}$$

Example 15 : Find the Laurent series of the function $f(z) = \frac{z}{(z+1)(z+2)}$ about $z = -2$.

[JNTU 2005S, June 2009 S(Set No.2)]

Solution : By partial fractions,

$$\begin{aligned}
 f(z) &= \frac{z}{(z+1)(z+2)} = z \left[\frac{1}{z+1} - \frac{1}{z+2} \right] = \frac{z}{z+1} - \frac{z}{z+2} \\
 &= \frac{(z+2)-2}{(z+2)-1} - \frac{(z+2)-2}{z+2} \\
 &= \frac{(z+2)-2}{-1-(z+2)} - 1 + \frac{2}{z+2} = \frac{2-(z+2)}{1-(z+2)} - 1 + \frac{2}{z+2} \\
 &= -1 + \frac{2}{z+2} + [2-(z+2)] [1-(z+2)]^{-1} \\
 &= -1 + \frac{2}{z+2} + [2-(z+2)][1+(z+2)+(z+2)^2+\dots] \text{ if } |z+2| < 1 \\
 &= -1 + \frac{2}{z+2} + 2 \left[1+(z+2)+(z+2)^2+\dots \right] - (z+2)[1+(z+2)+(z+2)^2+\dots] \\
 &= -1 + \frac{2}{z+2} + [2+2(z+2)+2(z+2)^2+\dots] - [(z+2)+(z+2)^2+\dots] \\
 &= \frac{2}{z+2} + 1+(z+2)+(z+2)^2+\dots
 \end{aligned}$$

Example 16 : (i) Find the Laurent expansion of $\frac{1}{z^2-4z+3}$, for

(a) $1 < |z| < 3$.

[JNTU April 2006, 2008S (Set No. 1)]

(b) $|z| < 1$ (c) $|z| > 3$

[JNTU (K) Nov. 2009 (Set No. 4)]

(ii) Expand $\frac{1}{(z+3)(z+1)}$ in the annular region between $|z|=1$ and $|z|=3$.

[JNTU (K) Nov. 2009 (Set No. 1)]

Solution : (i) By partial fractions,

$$\frac{1}{z^2-4z+3} = \frac{1}{(z-1)(z-3)} = \frac{1}{2} \left[\frac{1}{z-3} - \frac{1}{z-1} \right] \quad \dots (1)$$

(a) For $1 < |z| < 3$, we write (1) as

$$\frac{1}{z^2-4z+3} = \frac{1}{2} \cdot \frac{-1}{3 \left(1 - \frac{z}{3} \right)} + \frac{1}{2z \left(1 - \frac{1}{z} \right)} = -\frac{1}{6} \left(1 - \frac{z}{3} \right)^{-1} + \frac{1}{2z} \left(1 - \frac{1}{z} \right)^{-1}$$

$$\begin{aligned}
 &= -\frac{1}{6} \left(1 + \frac{z}{3} + \frac{z^2}{9} + \frac{z^3}{27} + \dots \right) + \frac{1}{2z} \left(1 + \frac{1}{z} + \frac{1}{z^2} + \frac{1}{z^3} + \dots \right) \\
 &= -\frac{1}{6} \left(1 + \frac{z}{3} + \frac{z^2}{9} + \frac{z^3}{27} + \dots \right) + \frac{1}{2} \left(\frac{1}{z} + \frac{1}{z^2} + \frac{1}{z^3} + \dots \right)
 \end{aligned}$$

which is a Laurent's series.

(b) If $|z| < 1$, then we write (1) as

$$\begin{aligned}
 \frac{1}{z^2 - 4z + 3} &= \frac{1}{2} \left[-\frac{1}{3} \left(1 - \frac{z}{3} \right)^{-1} + (1 - z)^{-1} \right] \\
 &= \frac{1}{2} \left[-\frac{1}{3} \left(1 + \frac{z}{3} + \frac{z^2}{9} + \frac{z^3}{27} + \dots \right) + (1 + z^2 + z^3 + \dots) \right]
 \end{aligned}$$

(c) If $|z| > 3$ then $|3/z| < 1$. So, we write (1) as

$$\begin{aligned}
 \frac{1}{z^2 - 4z + 3} &= \frac{1}{2} \left[\frac{1}{z \left(1 - \frac{3}{z} \right)} - \frac{1}{z \left(1 - \frac{1}{z} \right)} \right] \\
 &= \frac{1}{2z} \left[\left(1 - \frac{3}{z} \right)^{-1} - \left(1 - \frac{1}{z} \right)^{-1} \right] \\
 &= \frac{1}{2z} \left[\left(1 + \frac{3}{z} + \frac{9}{z^2} + \frac{27}{z^3} + \dots \right) - \left(1 + \frac{1}{z} + \frac{1}{z^2} + \frac{1}{z^3} + \dots \right) \right]
 \end{aligned}$$

(ii) By Partial fractions,

$$\frac{1}{(z+3)(z+1)} = \frac{1}{2} \left(\frac{1}{z+1} - \frac{1}{z+3} \right) \quad \dots (1)$$

For $1 < |z| < 3$, we write (1) as

$$\begin{aligned}
 \frac{1}{(z+3)(z+1)} &= \frac{1}{2} \cdot \frac{1}{z \left(1 + \frac{1}{z} \right)} - \frac{1}{2} \cdot \frac{1}{3} \cdot \frac{1}{\left(1 + \frac{z}{3} \right)} \\
 &= \frac{1}{2z} \left(1 + \frac{1}{z} \right)^{-1} - \frac{1}{6} \left(1 + \frac{z}{3} \right)^{-1} \\
 &= \frac{1}{2z} \left(1 - \frac{1}{z} + \frac{1}{z^2} - \frac{1}{z^3} + \dots \right) - \frac{1}{6} \left(1 - \frac{z}{3} + \frac{z^2}{9} - \frac{z^3}{27} + \dots \right) \\
 &= \frac{1}{2z} \left(\frac{1}{z} - \frac{1}{z^2} + \frac{1}{z^3} - \frac{1}{z^4} + \dots \right) - \frac{1}{6} \left(1 - \frac{z}{3} + \frac{z^2}{9} - \frac{z^3}{27} + \dots \right)
 \end{aligned}$$

which is the required Laurent series.

Example 17 : Express $f(z) = \frac{z}{(z-1)(z-3)}$ in a series of positive and negative powers of $(z-1)$.

[JNTU 2006 (Set No. 2)]

Solution : $f(z) = \frac{z}{2} \left[\frac{2}{(z-1)(z-3)} \right] = \frac{z}{2} \left[\frac{1}{z-3} - \frac{1}{z-1} \right] = \frac{1}{2} \left[\frac{z}{z-3} - \frac{z}{z-1} \right]$

Let $z-1 = w$. Then

$$\begin{aligned} f(1+w) &= \frac{1}{2} \left[\frac{1+w}{w-2} - \frac{1+w}{w} \right] = \frac{1}{2} \left[\frac{(w-2)+3}{w-2} - \frac{1+w}{w} \right] \\ &= \frac{1}{2} \left[\left(1 + \frac{3}{w-2} \right) - \left(1 + \frac{1}{w} \right) \right] \\ &= \frac{1}{2} \left[\frac{3}{w-2} - \frac{1}{w} \right] = \frac{3}{2} \cdot \frac{1}{w-2} - \frac{1}{2w} = \frac{3}{4 \left(\frac{w}{2} - 1 \right)} - \frac{1}{2w} \\ &= -\frac{3}{4} \left(1 - \frac{w}{2} \right)^{-1} - \frac{1}{2w} \\ &= -\frac{3}{4} \sum_{n=0}^{\infty} \left(\frac{w}{2} \right)^n - \frac{1}{2w} \quad \text{if } \frac{|w|}{2} < 1 \quad \text{i.e. if } |w| < 2 \end{aligned}$$

Hence $f(z) = -\frac{3}{4} \sum_{n=0}^{\infty} \left(\frac{z-1}{2} \right)^n - \frac{1}{2(z-1)} \quad \text{if } |z-1| < 2$

or $f(z) = -\frac{3}{4} \sum_{n=0}^{\infty} \frac{(z-1)^n}{2^n} - \frac{1}{2(z-1)} \quad \text{if } |z-1| < 2$

Example 18 : Find the Laurent series expansion of the function $\frac{z^2-1}{(z+2)(z+3)}$ if $2 < |z| < 3$.

[JNTU 2006 (Set No. 3)]

Solution : Let $f(z) = \frac{z^2-1}{(z+2)(z+3)}$

$$= 1 - \frac{5z+7}{(z+2)(z+3)} \quad [\text{by actual division}]$$

$$= 1 + \frac{3}{z+2} - \frac{8}{z+3} \quad (\text{Resolving into partial fractions})$$

Given $2 < |z| < 3 \Rightarrow \frac{2}{|z|} < 1$ and $\frac{|z|}{3} < 1$

$$\begin{aligned}
 \therefore f(z) &= 1 + \frac{3}{z} \left(1 + \frac{2}{z}\right)^{-1} - \frac{8}{3} \left(1 + \frac{z}{3}\right)^{-1} \\
 &= 1 + \frac{3}{z} \left[1 - \frac{2}{z} + \left(\frac{2}{z}\right)^2 - \left(\frac{2}{z}\right)^3 + \dots\right] - \frac{8}{3} \left[1 - \frac{z}{3} + \left(\frac{z}{3}\right)^2 - \left(\frac{z}{3}\right)^3 + \dots\right] \\
 &= \frac{-5}{3} + \frac{3}{z} - \frac{6}{z^2} + \frac{12}{z^3} - \frac{24}{z^4} + \dots + \frac{8z}{9} - \frac{8z^2}{27} + \frac{8z^3}{81} + \dots
 \end{aligned}$$

Example 19 : Expand $\frac{1}{(z^2+1)(z^2+2)}$ in positive and negative powers of z if $1 < |z| < \sqrt{2}$.

[JNTU 2007S, June 2009S (Set No. 4)]

Solution : By partial fractions, $\frac{1}{(z^2+1)(z^2+2)} = \frac{1}{z^2+1} - \frac{1}{z^2+2}$

To get a Laurent's expansion in the region $1 < |z| < \sqrt{2}$, we note that

$$\frac{1}{|z|} < 1 \text{ and } \left| \frac{z}{\sqrt{2}} \right| < 1 \text{ or } \left| \frac{z^2}{2} \right| < 1$$

$$\begin{aligned}
 \text{Hence } \frac{1}{(z^2+1)(z^2+2)} &= \frac{1}{z^2 \left(1 + \frac{1}{z^2}\right)} - \frac{1}{2 \left(1 + \frac{z^2}{2}\right)} \\
 &= \frac{1}{z^2} \left(1 + \frac{1}{z^2}\right)^{-1} - \frac{1}{2} \left(1 + \frac{z^2}{2}\right)^{-1} \\
 &= \frac{1}{z^2} \left(1 - \frac{1}{z^2} + \frac{1}{z^4} - \frac{1}{z^6} + \dots\right) - \frac{1}{2} \left(1 - \frac{z^2}{2} + \frac{z^4}{2^2} - \frac{z^6}{2^3} + \dots\right) \\
 &= \left(\frac{1}{z^2} - \frac{1}{z^4} + \frac{1}{z^6} - \frac{1}{z^8} + \dots\right) - \left(\frac{1}{2} - \frac{z^2}{2^2} + \frac{z^4}{2^3} - \frac{z^6}{2^4} + \dots\right) \\
 &= \left(\frac{1}{z^2} - \frac{1}{z^4} + \frac{1}{z^6} - \frac{1}{z^8} + \dots\right) + \left(-\frac{1}{2} + \frac{z^2}{2^2} - \frac{z^4}{2^3} + \frac{z^6}{2^4} - \dots\right) \\
 &= \sum_{n=0}^{\infty} (-1)^n \frac{1}{z^{2n+2}} + \sum_{n=0}^{\infty} (-1)^{n+1} \frac{z^{2n}}{2^{n+1}}
 \end{aligned}$$

This is the required Laurent's expansion and this is valid in $1 < |z| < \sqrt{2}$.

Example 20 : Expand $f(z) = \frac{(z-2)(z+2)}{(z+1)(z+4)}$ in the region

$$(a) \quad 1 < |z| < 4$$

$$(b) \quad |z| < 1$$

[JNTU 2007S, June 2009S (Set No.3)]

Solution : We have

$$f(z) = \frac{(z-2)(z+2)}{(z+1)(z+4)} = \frac{z^2-4}{z^2+5z+4} = 1 - \frac{5z-1}{(z+1)(z+4)}$$

Resolving into partial fractions,

$$f(z) = 1 - \left(\frac{-2}{z+1} + \frac{7}{z+4} \right) \quad \dots (1)$$

$$= 1 + \frac{2}{z+1} - \frac{7}{z+4}$$

(a) When $1 < |z| < 4$, then $\frac{1}{|z|} < 1$ and $\frac{|z|}{4} < 1$

Hence $f(z) = 1 + \frac{2}{z} \left(1 + \frac{1}{z} \right)^{-1} - \frac{7}{4} \left(1 + \frac{z}{4} \right)^{-1}$

$$= 1 + \frac{2}{z} \left[1 - \frac{1}{z} + \frac{1}{z^2} - \frac{1}{z^3} + \dots \right] - \frac{7}{4} \left[1 - \frac{z}{4} + \left(\frac{z}{4} \right)^2 - \left(\frac{z}{4} \right)^3 + \dots \right]$$

$$= -\frac{3}{4} + 2 \left[-\frac{1}{z} + \frac{1}{z^2} - \frac{1}{z^3} + \frac{1}{z^4} - \dots \right] - \frac{7}{4} \left[-\frac{z}{4} + \left(\frac{z}{4} \right)^2 - \left(\frac{z}{4} \right)^3 + \dots \right]$$

$$= -\frac{3}{4} + 2 \sum_{n=1}^{\infty} (-1)^n \frac{1}{z^n} - \frac{7}{4} \sum_{n=1}^{\infty} (-1)^n \left(\frac{z}{4} \right)^n$$

(b) When $|z| < 1$, then $\frac{|z|}{4} < 1$

Now equation (1) can be written as

$$f(z) = 1 + 2(1+z)^{-1} - \frac{7}{4} \left(1 + \frac{z}{4} \right)^{-1}$$

$$= 1 + 2(1 - z + z^2 - z^3 + \dots) - \frac{7}{4} \left[1 - \frac{z}{4} + \left(\frac{z}{4} \right)^2 - \left(\frac{z}{4} \right)^3 + \dots \right]$$

$$= \frac{5}{4} - (z - z^2 + z^3 - \dots) + \frac{7}{4} \left[\frac{z}{4} - \left(\frac{z}{4} \right)^2 + \left(\frac{z}{4} \right)^3 - \dots \right]$$

$$= \frac{5}{4} - \sum_{n=1}^{\infty} (-1)^{n+1} z^n + \frac{7}{4} \sum_{n=1}^{\infty} (-1)^{n+1} \left(\frac{z}{4} \right)^n$$

Example 21 : Find the Laurent series expansion of the function $\frac{z^2 - 1}{z^2 + 5z + 6}$ about $z = 0$ in the region $2 < |z| < 3$.

[JNTU Nov. 2008S (Set No.4)]

Solution : Let $f(z) = \frac{z^2 - 1}{z^2 + 5z + 6}$

$$\begin{aligned}
 &= 1 - \frac{5z+7}{z^2+5z+6} \quad [\text{By actual division}] \\
 &= 1 - \frac{5z+7}{(z+2)(z+3)} \\
 &= 1 + \frac{3}{z+2} - \frac{8}{z+3} \quad (\text{Resolving into partial fractions})
 \end{aligned}$$

Given $|z| > 2 \Rightarrow \frac{2}{|z|} < 1$ and $|z| < 3 \Rightarrow \frac{|z|}{3} < 1$

$$\begin{aligned}
 \therefore f(z) &= 1 + \frac{3}{z\left(1+\frac{2}{z}\right)} - \frac{8}{3\left(1+\frac{z}{3}\right)} \\
 &= 1 + \frac{3}{z}\left(1+\frac{2}{z}\right)^{-1} - \frac{8}{3}\left(1+\frac{z}{3}\right)^{-1} \\
 &= 1 + \frac{3}{z}\left[1 - \frac{2}{z} + \left(\frac{2}{z}\right)^2 - \left(\frac{2}{z}\right)^3 + \dots\right] \\
 &\quad - \frac{8}{3}\left[1 - \left(\frac{z}{3}\right) + \left(\frac{z}{3}\right)^2 - \left(\frac{z}{3}\right)^3 + \dots\right] \\
 &= 1 + \frac{3}{z} \sum_{n=0}^{\infty} (-1)^n \left(\frac{2}{z}\right)^n - \frac{8}{3} \sum_{n=0}^{\infty} (-1)^n \left(\frac{z}{3}\right)^n
 \end{aligned}$$

which is the required laurent series.

Example 22 : Expand $f(z) = \frac{1+2z}{z^2+z^3}$ in a series of positive and negative powers of z .

[JNTU Nov. 2008 (Set No.1)]

Solution : We have $f(z) = \frac{1+2z}{z^2+z^3} = \frac{1+2z}{z^2(1+z)}$

$$= \frac{1}{z^2} \left[\frac{1+2z}{1+z} \right] = \frac{1}{z^2} \left[\frac{2+2z-1}{1+z} \right] = \frac{1}{z^2} \left[2 - \frac{1}{1+z} \right]$$

This function $f(z)$ is not defined at $z = 0$. $z = 0$ is a singular point. But for the part, the quantity within the square brackets can be expanded in non-negative powers of z . Using the standard

expansion of $\frac{1}{1+z}$, we have

$$\begin{aligned}
 f(z) &= \frac{1}{z^2} \left[2 - \{1 - z + z^2 - z^3 + \dots\} \right] \\
 &= \frac{1}{z^2} \left[1 + z - z^2 + z^3 - z^4 + \dots \right] \\
 &= \frac{1}{z^2} + \frac{1}{z} - 1 + z - z^2 + z^3 + \dots
 \end{aligned}$$

and this expansion is valid for $|z| < 1$ except for $z = 0$. i.e., It is valid for $0 < |z| < 1$.

Example 23 : Represent the function $f(z) = \frac{z}{(z-1)(z-3)}$ by a series of positive and negative powers of $(z-1)$ which converges to $f(z)$ when $0 < |z-1| < 2$.

Sol. Since we want the expansion of $f(z)$ in powers of $z-1$,

put $z-1 = w$, i.e., $z = 1 + w$.

$$\therefore f(z) = \frac{1+w}{w(w-2)}$$

$$\text{Write } \frac{1+w}{w(w-2)} = \frac{A}{w} + \frac{B}{w-2}$$

$$\Rightarrow A(w-2) + Bw = 1+w$$

$$\text{Put } w = 0 \Rightarrow -2A = 1 \quad \therefore A = -1/2$$

$$\text{Put } w = 2; \Rightarrow 2B = 3 \quad \therefore B = 3/2$$

$$\therefore \frac{1+w}{w(w-2)} = -\frac{1}{2w} + \frac{3}{2(w-2)}$$

$$= -\frac{1}{2w} - \frac{3}{2(2-w)} = -\frac{1}{2w} - \frac{3}{4\left(1-\frac{w}{2}\right)}$$

$$= -\frac{1}{2w} - \frac{3}{4} \left[1 + \frac{w}{2} + \left(\frac{w}{2}\right)^2 + \left(\frac{w}{2}\right)^3 + \dots \right] \text{ if } w \neq 0 \text{ and } \left|\frac{w}{2}\right| < 1$$

$$= -\frac{1}{2(z-1)} - \frac{3}{4} \left[1 + \frac{z-1}{2} + \left(\frac{z-1}{2}\right)^2 + \left(\frac{z-1}{2}\right)^3 + \dots \right] \text{ if } z-1 \neq 0 \text{ and } |z-1| < 2$$

$$= -\frac{1}{2(z-1)} - \frac{3}{4} \sum_{n=0}^{\infty} \left(\frac{z-1}{2}\right)^n \text{ if } 0 < |z-1| < 2.$$

EXERCISE

1. Find the Taylor's series expansion of $\sin z$ around $z = \pi/2$.

2. Find the Taylor's series expansion of e^z about $z = 3$.

$$\left[\text{Ans. } e^3 \sum_{n=0}^{\infty} \frac{(z-3)^n}{n!} \right]$$

3. Expand $f(z) = \cos z$ in Taylor's series about $z = \frac{\pi}{4}$.

$$\left[\text{Ans. } \cos z = -\left(z - \frac{\pi}{2}\right) + \frac{1}{3!} \left(z - \frac{\pi}{2}\right)^3 - \frac{1}{5!} \left(z - \frac{\pi}{2}\right)^5 + \dots \right]$$

4. Find the Maclaurin's series expansion of $f(z)$ given below :

(a) e^z (b) $\cos az$ (c) $\cosh z$ (d) $\sinh z$

(e) $\log(1+z)$ (f) $(1+z)^n$ where $n = -1, -2$.

5. Expand $f(z) = \frac{1}{(z-1)(z-2)}$ in the region (a) $|z| < 1$ (b) $1 < |z| < 2$ (c) $|z| > 2$.
6. Find the Laurent's series of

(a) $\frac{e^{2z}}{(z-1)^2}$ about $z = 1$.

(b) $\frac{z}{(z-1)(z-2)}$ about $z = 0$ in the three different ways.

(c) $\frac{z}{(z-1)(z-3)}$ about $z = 3$ in the two different ways.

7. Find the Laurent's series of $\frac{z}{(z^2-1)(z^2+4)}$ for

(a) $|z| < 1$ (b) $1 < |z| < 2$ (c) $|z| > 2$.

8. Find the Laurent's series of $\frac{1}{z^2(z+i)}$ about $(z = -i)$ and hence find the residue at $z = i$.

9. Expand $f(z) = \frac{1}{(z+1)(z+2)}$ in Laurent's series valid for

(a) $1 < |z| < 3$ (b) $|z| > 3$

(c) $0 < |z+1| < 2$ (d) $|z| > 1$

10. Expand $\frac{1}{(z^2+1)(z^2+2)}$ in positive and negative powers of z , if

(i) $|z| < 1$

(ii) $|z| > \sqrt{2}$

[Ans. (i) $\sum_{n=0}^{\infty} (-1)^n \left[1 - \frac{1}{2^{n+1}}\right] (z^2)^n$ (ii) $\sum_{n=0}^{\infty} (-1)^n z^{-2n-2} (1 - 2^n)$]

11. Expand $f(z) = \frac{(z-2)(z+2)}{(z+1)(z+4)}$ in the region $|z| > 4$.

[Ans. $1 - \sum_{n=1}^{\infty} (-1)^n \left\{ \frac{1}{z^{n+1}} + \left(\frac{4}{z}\right)^{n+1} \right\}$]

5.6. DEFINITIONS.

(1) Zero of an Analytic Function.

A zero of an analytic function $f(z)$ is a value of z such that $f(z) = 0$. Particularly, a point a is called a zero of an analytic function $f(z)$ if $f(a) = 0$.

Zero of m th order. If an analytic function $f(z)$ can be expressed in the form

$$f(z) = (z-a)^m \phi(z)$$

where $\phi(z)$ is analytic and $\phi(a) \neq 0$, then $z = a$ is called zero of m th order of the function $f(z)$. Simple zero is a zero of order one.

For example, if

(i) $f(z) = (z-1)^3$, then $z = 1$ is a zero of order 3 of $f(z)$

(ii) $f(z) = \frac{1}{1-z}$, then $z = \infty$ is a simple zero of $f(z)$.

(iii) $f(z) = \sin z$, then $z = 0, \pm \pi, \pm 2\pi, \dots$ are simple zeros of $f(z)$.

(iv) $f(z) = e^{\tan z}$ has no zeros since $e^z \neq 0$ for any z .

(2) Singular Points.

A singular point (or singularity) of a function $f(z)$ is the point at which the function $f(z)$ ceases to be analytic.

Different types of Singularities.

(i) **Isolated Singularity.** A point $z = a$ is called an Isolated singularity of an analytic function $f(z)$, if

(a) $f(z)$ is not analytic at the point $z = a$ ✓

(b) $f(z)$ is analytic in the deleted neighbourhood of $z = a$, i.e. there exists a neighbourhood of the point $z = a$ which contains no other singularity.

For example,

if $f(z) = \frac{e^z}{z^2 + 1}$, then $z = \pm i$ are two isolated singular points of $f(z)$.

if $f(z) = \frac{2}{\sin z}$, 

then $z = \pm \pi, \pm 2\pi, \pm 3\pi, \dots$ are infinite number of isolated singular points of $f(z)$.

(3) Poles of an Analytic Function.

If $z = a$ is an isolated singular point of an analytic function $f(z)$, then $f(z)$ can be expanded in Laurent's series about the point $z = a$,

$$\text{i.e. } f(z) = \sum_{n=0}^{\infty} a_n (z-a)^n + \sum_{n=1}^{\infty} \frac{b_n}{(z-a)^n} \quad \dots (1)$$

The series of negative integral powers of $(z-a)$ namely $\sum_{n=1}^{\infty} \frac{b_n}{(z-a)^n}$ is known as the "principal part" of the Laurent's series of $f(z)$.

If the principal part contains a finite number of terms, say m , (i.e. $b_n = 0 \quad \forall n$ such that $n > m$), then the singular point $z = a$ is called a **pole** of order m of $f(z)$.

Simple Pole is a pole of order one.

For example,

if $f(z) = \frac{z^2}{(z-1)(z+2)^2}$, then $z = 1$ is a simple pole and $z = -2$ is a pole of order 2.

An Alternative Definition. If there exists a positive integer n such that $\lim_{z \rightarrow a} (z-a)^n f(z) = A \neq 0$ then $z = a$ is called a pole of order n .

(4) Essential Singularity.

If the principal part of $f(z)$ contains an infinite number of terms, i.e. the series $\sum_{n=1}^{\infty} \frac{b_n}{(z-a)^n}$ contains an infinite number of terms, then the point $z = a$ is called *Essential Singularity* of $f(z)$.

For example, $z = 0$ is an essential singularity of $e^{1/z}$, since the principal part of $e^{1/z}$ contains infinite number of terms containing negative powers of $(z - 0)$.

An Alternative Definition.

If there exists no finite value of n such that $\lim_{z \rightarrow 0} (z-a)^n f(z) = \text{finite non-zero constant}$, then $z = a$ is called essential singularity.

(5) Removable Singularity.

If the principal part of $f(z)$ contains no term i.e. if $b_n = 0 \forall n$, then the singularity $z = a$ is called removable singularity of $f(z)$. In this case

$$f(z) = \sum_{n=0}^{\infty} a_n (z-a)^n$$

For example, if $f(z) = \frac{1 - \cos z}{z}$, then $z = 0$ is a removable singularity

An Alternative Definition.

If a single-valued function $f(z)$ is not defined at $z = a$ but $\lim_{z \rightarrow a} f(z)$ exists, then the singularity $z = a$ is known as a removable singularity.

For example, if $f(z) = \frac{\sin z}{z}$ then $f(0)$ is not defined but $\lim_{z \rightarrow 0} \frac{\sin z}{z} = 1$. So $z = 0$ is called

Removable singularity.

(6) Singularities at Infinity.

Taking $z = \frac{1}{t}$ in $f(z)$ we obtain $f\left(\frac{1}{t}\right) = F(t)$. Then the nature of the singularity at $z = \infty$ is defined to be the same as that of $F(t)$ at $t = 0$.

For example, (i) $f(z) = z^3$ has a pole of order 3 at $z = \infty$, since $f\left(\frac{1}{t}\right) = \frac{1}{t^3}$ has a pole of order 3 at $t = 0$ (ii) $f(z) = e^z$ has an essential singularity at $z = \infty$, since $f\left(\frac{1}{t}\right) = e^{1/t}$ has an essential singularity at $t = 0$.

$$\begin{aligned} & \frac{1}{10} \\ & \frac{2 + \frac{2}{2}}{2} \\ & = 1 \end{aligned}$$